

Big Data analysis and dimensionality reduction to predict price trends in the Brazilian electricity market considering interdisciplinary phenomena

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This manuscript complements paper ID 9771 and presents all the results and figures produced for the dimensionality reduction of the database. The information derived from the feature engineering process is crucial for predicting price trends in the Brazilian electricity market, enabling traders to make well-informed decisions.

Figure 4 in the paper presents the methodological procedure, which begins with the construction of the product matrix ($t_x [MM, YY]$). Depending on the level of risk, traders can start trading a specific product up to one year in advance. Once a product is delivered, it ceases to exist. As a result, the historical time series of a product in the electricity market is relatively short compared to other commodities (e.g., steel and oil). To address this limitation, the product matrix, $t_x (MM, YY)$, shown in Eq. (3) in the paper, is manipulated to form a roll-off structure, as described in Eq. (6).

$$\begin{aligned}
 & t_{M+i \Rightarrow i=0}^{roll-off} [t_x (MM, YY)] = \\
 & \{t_x \mid t_x \in MM \equiv mm + i \mid \forall x=0, \dots, T\} \\
 & = \left[\begin{array}{cc}
 \begin{array}{l}
 d = d(yy, mm, dd) + 0 \\
 \{d \mid d(yy, mm, dd) + 0 \in mm \equiv MM - 0 \Leftrightarrow t_0(MM, YY)\} \\
 d = d(yy, mm, dd) + 1 \\
 \{d \mid d(yy, mm, dd) + 1 \in mm \equiv MM - 0 \Leftrightarrow t_0(MM, YY)\} \\
 d = d(yy, mm, dd) + 2 \\
 \{d \mid d(yy, mm, dd) + 2 \in mm \equiv MM - 0 \Leftrightarrow t_0(MM, YY)\} \\
 \vdots \\
 d = d(yy, mm, dd) + j \\
 \{d \mid d(yy, mm, dd) + j \in mm \equiv MM - 0 \Leftrightarrow t_0(MM, YY)\} \forall j=0, \dots, J \\
 d = d(yy, mm, dd) + k \\
 \{d \mid d(yy, mm, dd) + k \in mm \equiv MM - 0 \Leftrightarrow t_1(MM, YY)\} \forall k=J+1, \dots, K \\
 \vdots \\
 d = d(yy, mm, dd) + l \\
 \{d \mid d(yy, mm, dd) + l \in mm \equiv MM - 0 \Leftrightarrow t_2(MM, YY)\} \forall l=K+1, \dots, L \\
 \vdots \\
 d = d(yy, mm, dd) + z \\
 \{d \mid d(yy, mm, dd) + z \in mm \equiv MM - i \Leftrightarrow t_x(MM, YY)\} \forall z=L+1, \dots, Z
 \end{array}
 & \begin{array}{l}
 s_{t_0, d+0} \quad q_{t_0, d+0} \\
 s_{t_0, d+1} \quad q_{t_0, d+1} \\
 s_{t_0, d+2} \quad q_{t_0, d+2} \\
 \vdots \\
 s_{t_0, d+j} \quad q_{t_0, d+j} \\
 s_{t_1, d+k} \quad q_{t_1, d+k} \\
 \vdots \\
 s_{t_2, d+l} \quad q_{t_2, d+l} \\
 \vdots \\
 s_{t_x, d+z} \quad q_{t_x, d+z}
 \end{array}
 \end{array} \right] \quad (6)
 \end{aligned}$$

The roll-off matrix for $M+i$, where $i = 0$, is constructed for all products, T , starting from the initial reference product, $t_x (MM, YY)$, where $t_x : x = 0, \dots, T$. These products are progressively grouped beginning from the first record of

the historical series, $d = d(yy, mm, dd) + 0$, with the trading date, d , coinciding with the month, $mm = MM - 0$, in advance, $i = 0$, during which the initial benchmark product, $t_0 (MM, YY)$, is traded.

Subsequently, the trading dates, $d = d(yy, mm, dd) + j$, are limited to the set of trades that fall within the calendar month in which the product, $t_0 (MM, YY)$, is traded, $mm = MM - 0$, progressing from, $j = 0, \dots, J$, until the last trading date, $d = d(yy, mm, dd) + J$, of that month. For the next product, $t_1 (MM, YY)$, the trading dates, $d = d(yy, mm, dd) + k$, also fall within the same month, $mm = MM - 0$, $i = 0$, in which the reference product, $t_1 (MM, YY)$, is traded and progress from, $k = J+1, \dots, K$. Similarly, for the subsequent reference product, $t_2 (MM, YY)$, the trading dates are, $d = d(yy, mm, dd) + l : mm = MM - 0$, where $l = K+1, \dots, L$. This procedure continues until the last reference product, $t_x (MM, YY)$, with trading dates, $d = d(yy, mm, dd) + z : mm = MM - 0$, where $z = L+1, \dots, Z$ up to the end of the historical series at the most recent recorded date. In this way, the historical series for the Forward Curve projects a near future based on the recent past.

For example, in this case, the objective is to capture the trading behavior of economic agents for a product with a lead time before delivery, specifically for $M+i$, where $i = 1$. Taking the reference, $t_0 = APR2024$, the information series used to construct the Forward Curve, $M+1$, will consist of the historical series of the observed variables for each product within the same anticipation period, $i = 1$. Thus, the series is composed of prices and volumes (observed variables) traded in: March, $d (yy, mm, dd) : mm = MM - 1$, for the product $APR = t_0 (MM, YY)$; April for product $MAI = t_1 (MM, YY)$; May for product $JUN = t_2 (MM, YY)$; and June for product $JUL = t_3 (MM, YY)$; and so on.

The next step is to incorporate the behavior of the features into the product matrix, forming an extensive database. According to the information matrix for each product in Eq. (6), the features are organized in matrix (7) by computing each feature, $f = 1, \dots, F$, for each day, d .

$$\begin{bmatrix} d = d(yy, mm, dd) + 0 & S_{h=H-j,t,d} & q_{h=H-j,t,d} & f_{1,d} & \dots & f_{F,d} \\ \{d|d \in d(yy, mm, dd) + j\} \forall j=0, \dots, J & \{h|h \in H-j\} \forall j=0, \dots, J & & & & \\ d = d(yy, mm, dd) + 1 & S_{h=H-1,t,d} & q_{h=H-1,t,d} & & & \\ d = d(yy, mm, dd) + 2 & S_{h=H-2,t,d} & q_{h=H-2,t,d} & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & & & \\ d = d(yy, mm, dd) + j & S_{h=0,t,d} & q_{h=0,t,d} & f_{1,d} & \dots & f_{F,d} \\ \{d:d=d(yy, mm, dd) + j \rightarrow d(YY, MM, DD)\} & \{h:h=H-j \rightarrow h=0\} & & & & \end{bmatrix} \quad (7)$$

In (7), the 13 features are mathematically organized into five groups. The Hydrological group consists of variations in historical Affluent Natural Energy (ENA), with a shift, sh , of 0, 30, 60, 90, and 120 days for each Simple Moving Average (SMA) of 5, 7, 15, 30, 60, and 90 days (TABLE I).

TABLE I. HYDROLOGICAL FEATURES

Feature	Description
$ENA_{hist,d}(sh)$ $\{sh sh \in \{0,30,60,90,120\}\}$	Historical Affluent Natural Energy (ENA) with shift, sh , of 0, 30, 60, 90, and 120 days
$ENA_{hist-clim,d}(sh)$ $\{sh sh \in \{0,30,60,90,120\}\}$	Historical Affluent Natural Energy (ENA) subtracted from climatology with shift, sh , of 0, 30, 60, 90, and 120 days
$ENA_{hist,d}(sh, SMA)$ $\{sh sh \in \{0,30,60,90,120\}\}$ $\{SMA SMA \in \{5,7,15,30,60,90\}\}$	Historical Affluent Natural Energy (ENA) for each Simple Moving Average (SMA) of 5, 7, 15, 30, 60, and 90 days with shift, sh , of 0, 30, 60, 90, and 120 days

Note: The raw ENA data was obtained via an Application Programming Interface (API) through a protocol with the ONS [1], in collaboration with the company AMPERE Consultoria [2].

The Energetic group comprises the Marginal Operating Cost (CMO) of the four subsystems, sb , (South, Southeast/Central-West, Northeast and North) of the SIN; the capacity and size of the SIN transmission network; the Electricity Load Projection (Proj_C); and the Average Stored Energy (EARm) of the four subsystems, sb , (South, Southeast/Central-West, Northeast and North) and the SIN (TABLE II).

TABLE II. ENERGETIC FEATURES

Feature	Description
$CMO_d(sb)$ $\{sb \mid sb \in (S, SE, CW, NE, N)\}$	Marginal Cost of Operation (CMO) of each subsystem, s , South, Southeast/Central-West, Northeast, North.
$Cap_{LT,d}(SIN)$	Total capacity of SIN transmission lines [MVA]
$Siz_{LT,d}(SIN)$	Total size of SIN transmission lines [Km]
$EARM_d(sb)$ $\{sb \mid sb \in (S, SE, CW, NE, N, SIN)\}$	Average Stored Energy (EARM) of each subsystem, sb , South, Southeast/Central-West, Northeast, North, and SIN.
$Proj_C_d(M+i)$ $\{i \mid i \in (0, 1, 2, 3, 4)\}$	Electricity Load Projection, $Proj_C$, for $M+i$, in advance, $i = 0, \dots, 4$.

Note: The raw data for CMO , Cap_{LT} , Siz_{LT} , $EARM$, and $Proj_C$ were obtained via an Application Programming Interface (API) through a protocol with the ONS [1].

The Climatic group comprises climatology (Clim), the El Niño Mean Ocean Index (ONIm), and the Tropical North Atlantic Ocean Water Temperature Index (TNA), see TABLE III. The climatology refers to the country's climatological normal (the expected behavior for each year) or the long-term historical average of the ENA for the period analyzed.

TABLE III. CLIMATIC FEATURES

Feature	Description
$Clim_d(sh)$ $\{sh \mid sh \in (0, 30, 60, 90, 120)\}$	Climatology (Clim) with shift, sh , of 0, 30, 60, 90, and 120 days
ONI_d	El Niño Mean Ocean Index (ONIm)
TNA_d	Tropical North Atlantic Ocean Water Temperature Index (TNA)

Note: The raw $Clim$ data was obtained via an Application Programming Interface (API) through a protocol with the ONS [1], and the raw ONI and TNA data were obtained from AMPERE Consultoria [2].

The Economic group comprises the Gross Domestic Product (GDP) index, the Economic Activity Index (IBC), and the Volume Weighted Average Price (TABLE IV). The Volume Weighted Average Price (VWAP) is defined by Eq. **Erro! Fonte de referência não encontrada.** in the paper.

TABLE IV. ECONOMIC FEATURES

Feature	Description
GDP_d	Gross Domestic Product (GDP)
IBC_d	Economic Activity Index (IBC)
$s_{h,t,d}^{VWAP}$	Volume Weighted Average Price (VWAP)

Note: The raw GDP and IBC data were obtained via an Application Programming Interface (API) through a protocol with the Brazilian Central Bank [3]

The Geopolitical group comprises the Geopolitical Risk Index (GPRD), TABLE V. This feature was chosen as a qualitative parameter due to the political-regulatory influence on commodity prices.

TABLE V. GEOPOLITICAL FEATURES

Feature	Description
$GPRD_d$	Geopolitical Risk Index (GPRD)

Note: The raw GPRD data was obtained via an Application Programming Interface (API) from Economic Policy Uncertainty [4]

After forming the initial database and calculating the Spearman Correlation (SC) and Mutual Information (MI) indexes, the convergence analysis is performed based on the intensity of the SC index (Table VI) and the MI index (Table VII).

TABLE VI. SPEARMAN CORRELATION CLASSIFICATION STANDARDS

Classification Standards	Correlation Degree
$\rho = 0$	No Correlation
$0 < \rho \leq 0.20$	Indifferent
$0.20 < \rho \leq 0.40$	Weak
$0.40 < \rho \leq 0.60$	Moderate
$0.60 < \rho \leq 0.80$	Strong
$0.80 < \rho < 1.00$	Absolute
1	Monotonic Correlation

Source: adapted from [5]

The SC index, ρ , is determined as in Eq. (4) in the paper, and the evaluation is performed using the descriptive statistics of the qualitative scale (correlation degree) over the period, since a value of ρ is obtained for each feature on each day.

TABLE VII. MUTUAL INFORMATION CLASSIFICATION STANDARDS

Classification Standards	Correlation Degree
MI = 0	No Dependency
$0 < MI \leq P20 (MI)$	Very Low
$P20 (MI) < MI \leq P40 (MI)$	Low
$P40 (MI) < MI \leq P60 (MI)$	Moderate
$P60 (MI) < MI \leq P80 (MI)$	High
$P80 (MI) < MI < \max (MI)$	Very High
MI = max (MI)	Dependency Extreme

The MI index is determined as in Eq. (5) in the paper, and the evaluation is conducted through the descriptive statistics of the qualitative scale (correlation degree) over the period, since an MI value is obtained for each feature on each day. The scale is discretized according to the P20, P40, P60, and P80 percentiles of the IM results.

The factored database is determined after validating the SC and MI evaluations in conjunction with the trading industry, taking into account regulatory events as well as unexpected conditions in the operation of the electrical power system during the analyzed period. The validation procedure is carried out through out-of-sample evaluations using specific case studies selected by the trading sector.

The results obtained can be visualized in the figures generated by feature engineering, using SC and MI. The figures are presented sequentially below.

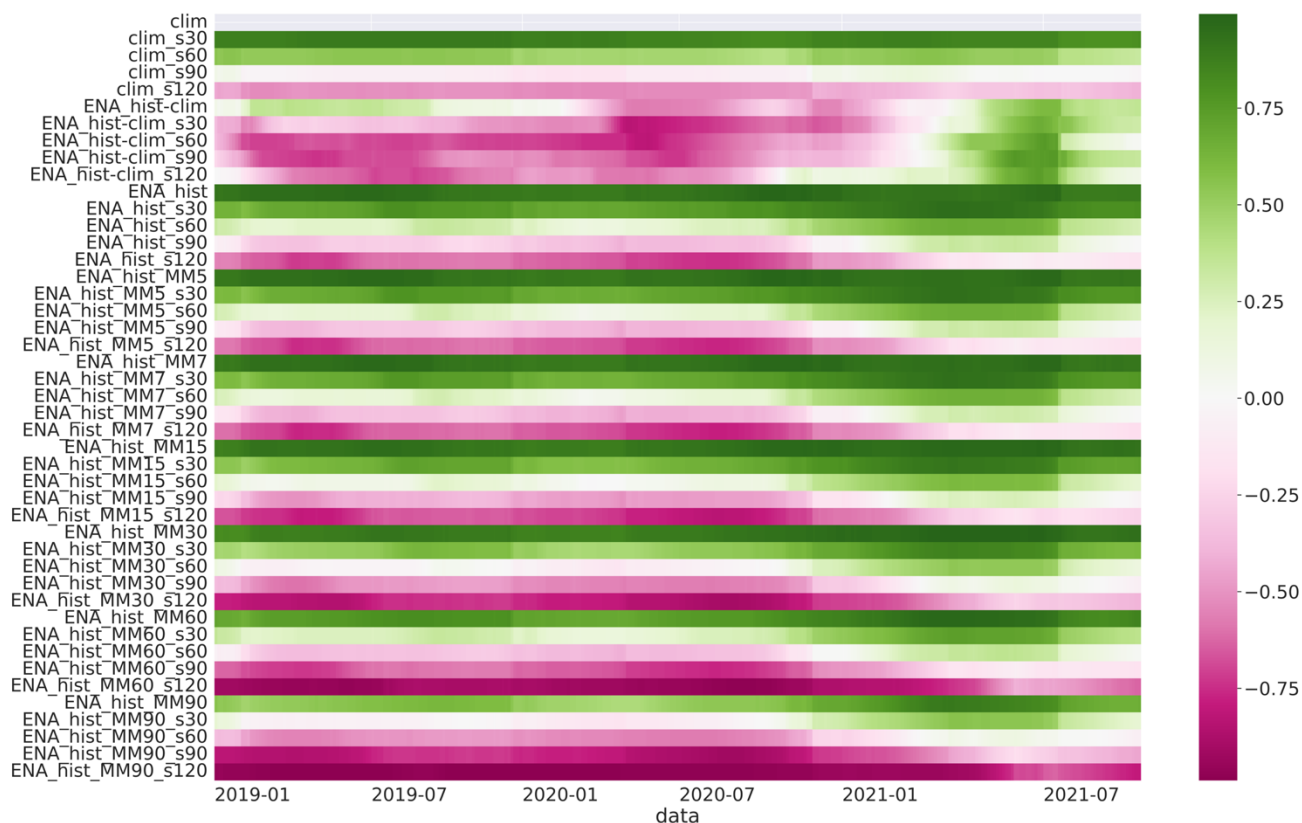


Fig. 1. Spearman correlation for the Climatology (Clim) feature.

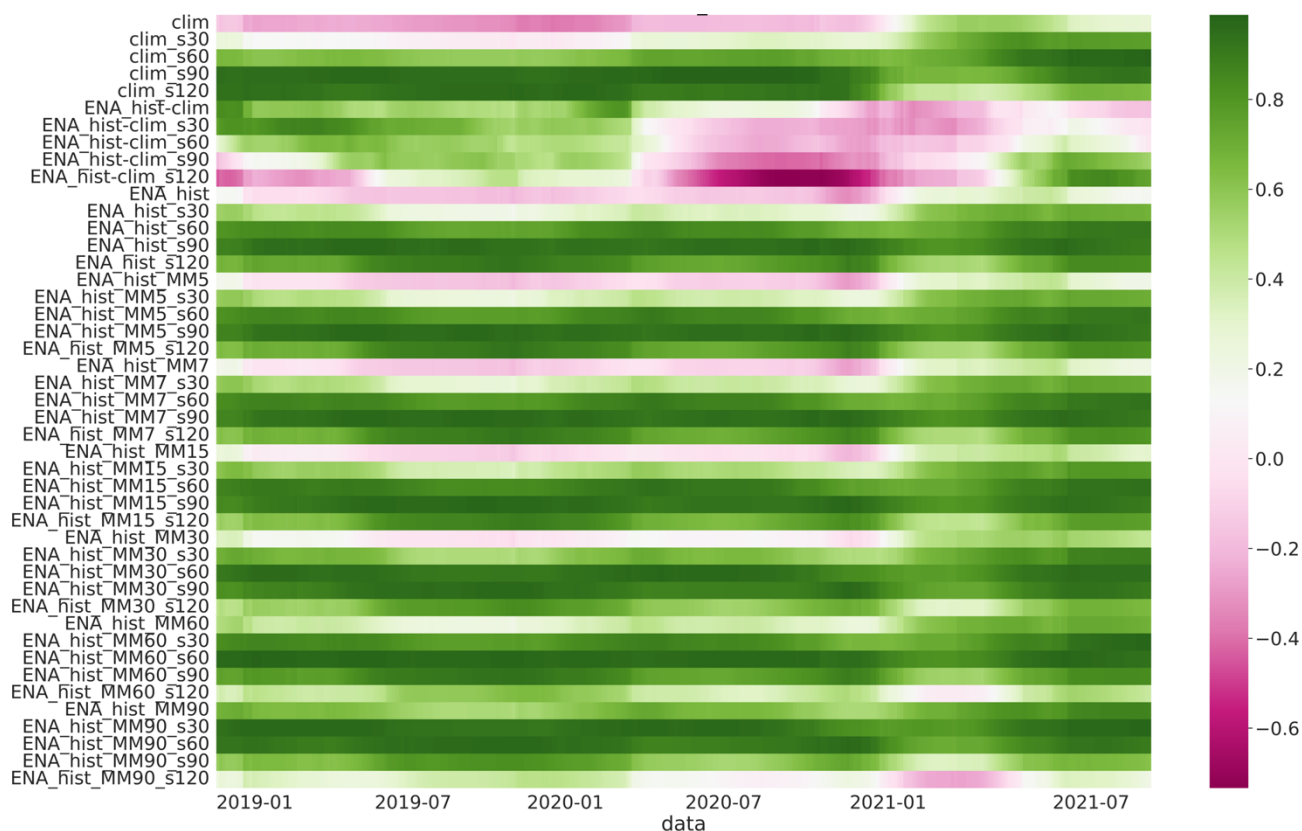


Fig. 2. Spearman correlation for the Average Stored Energy (EARm) feature of the North subsystem.



Fig. 3. Spearman correlation for the Average Stored Energy (EARm) feature of the Northeast subsystem.

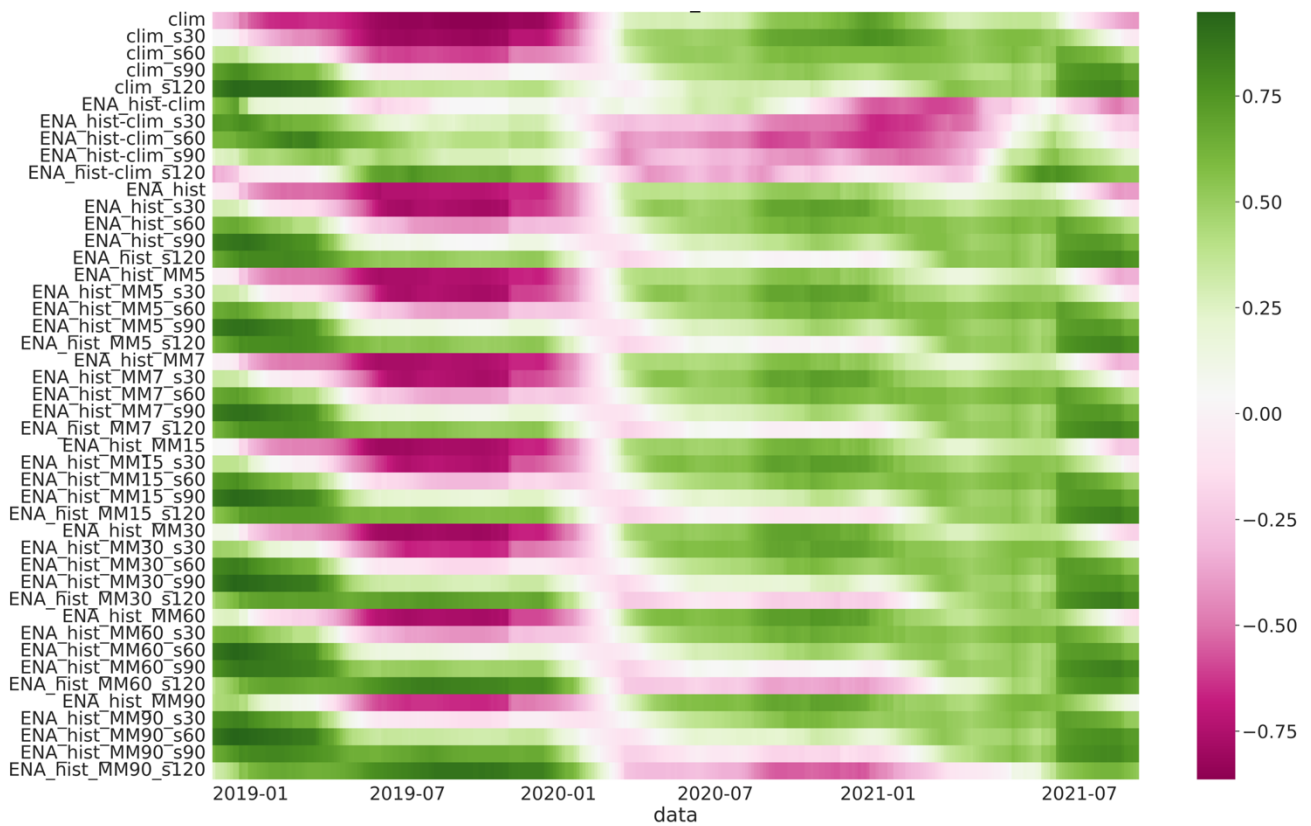


Fig. 4. Spearman correlation for the Average Stored Energy (EARm) feature of the South subsystem.



Fig. 5. Spearman correlation for the Average Stored Energy (EARm) feature of the Southeast/Central-West subsystem.



Fig. 6. Spearman correlation for the Average Stored Energy (EARm) feature of the National Interconnected System (SIN).

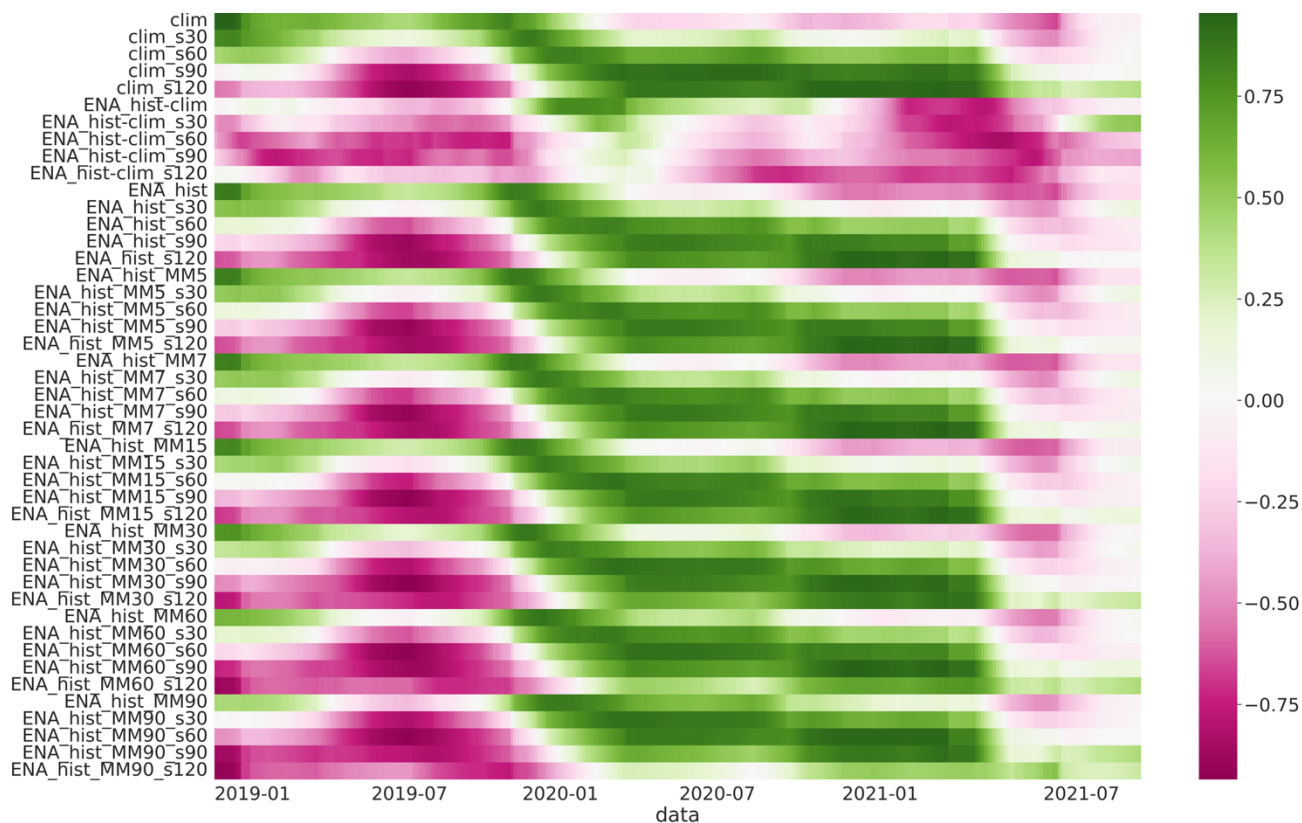


Fig. 7. Spearman correlation for the El Niño Mean Ocean Index (ONIm) feature.

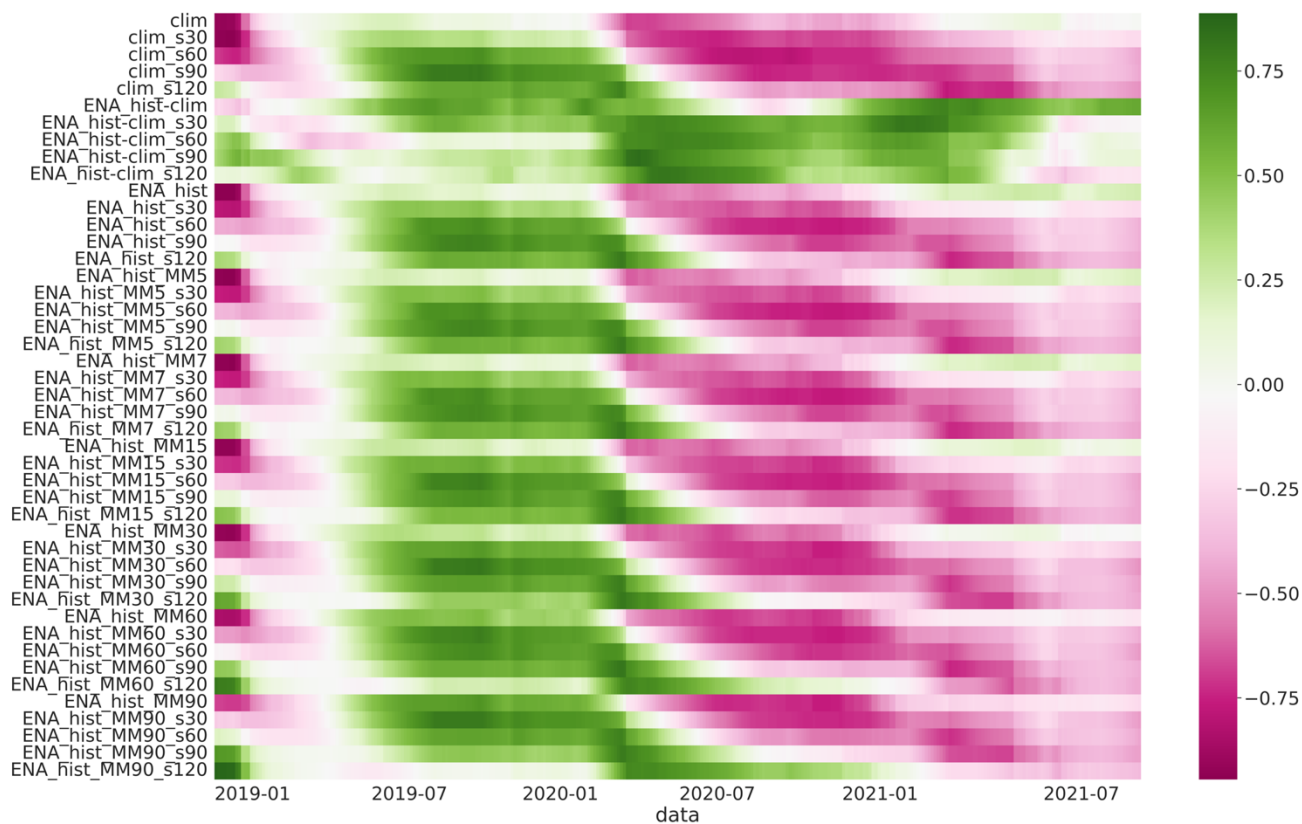


Fig. 8. Spearman correlation for the Tropical North Atlantic Ocean Water Temperature Index (TNA) feature.



Fig. 9. Spearman correlation for the Electricity Load Projection feature for the product M+0.

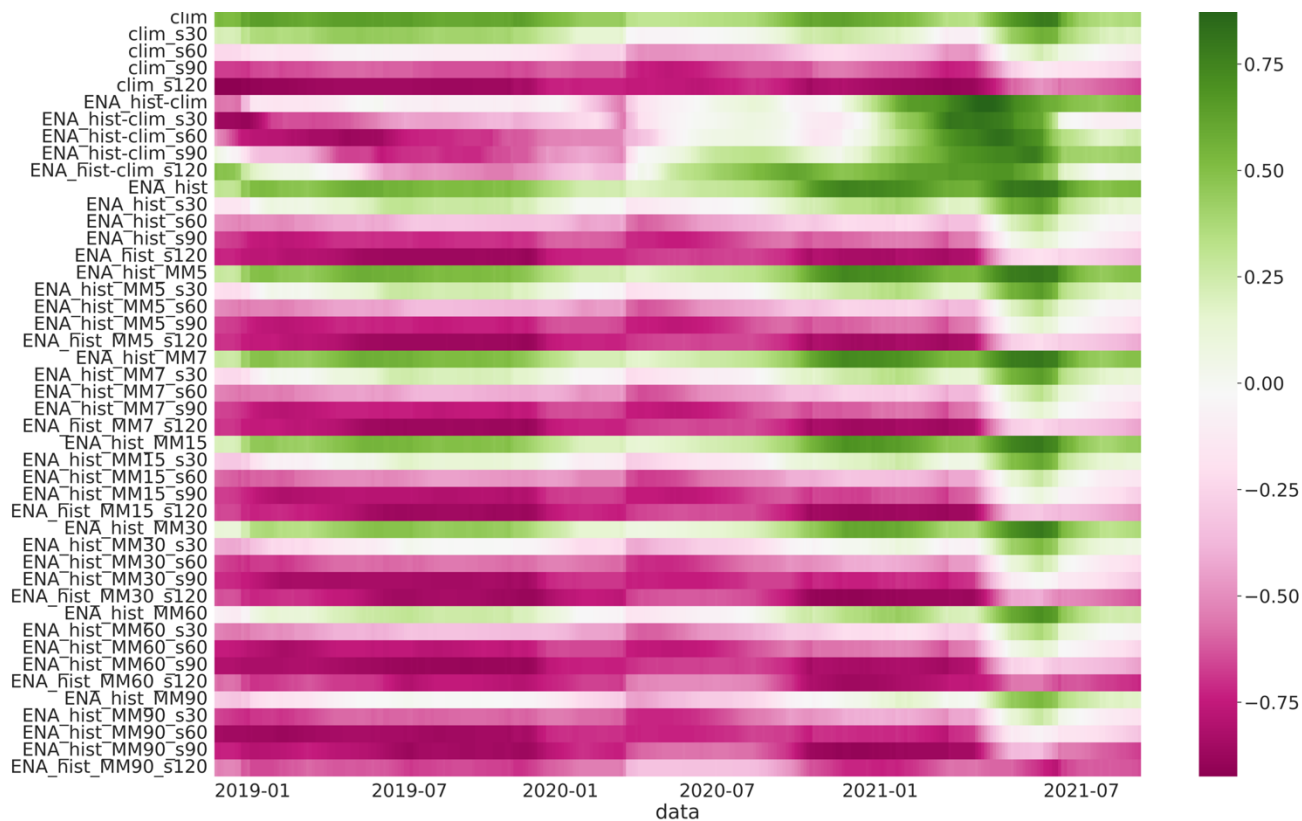


Fig. 10. Spearman correlation for the Electricity Load Projection feature or the product M+1.



Fig. 11. Spearman correlation for the Electricity Load Projection feature for the product M+2.



Fig. 12. Spearman correlation for the Electricity Load Projection feature for the product M+3.



Fig. 13. Spearman correlation for the Electricity Load Projection feature for the product M+4.

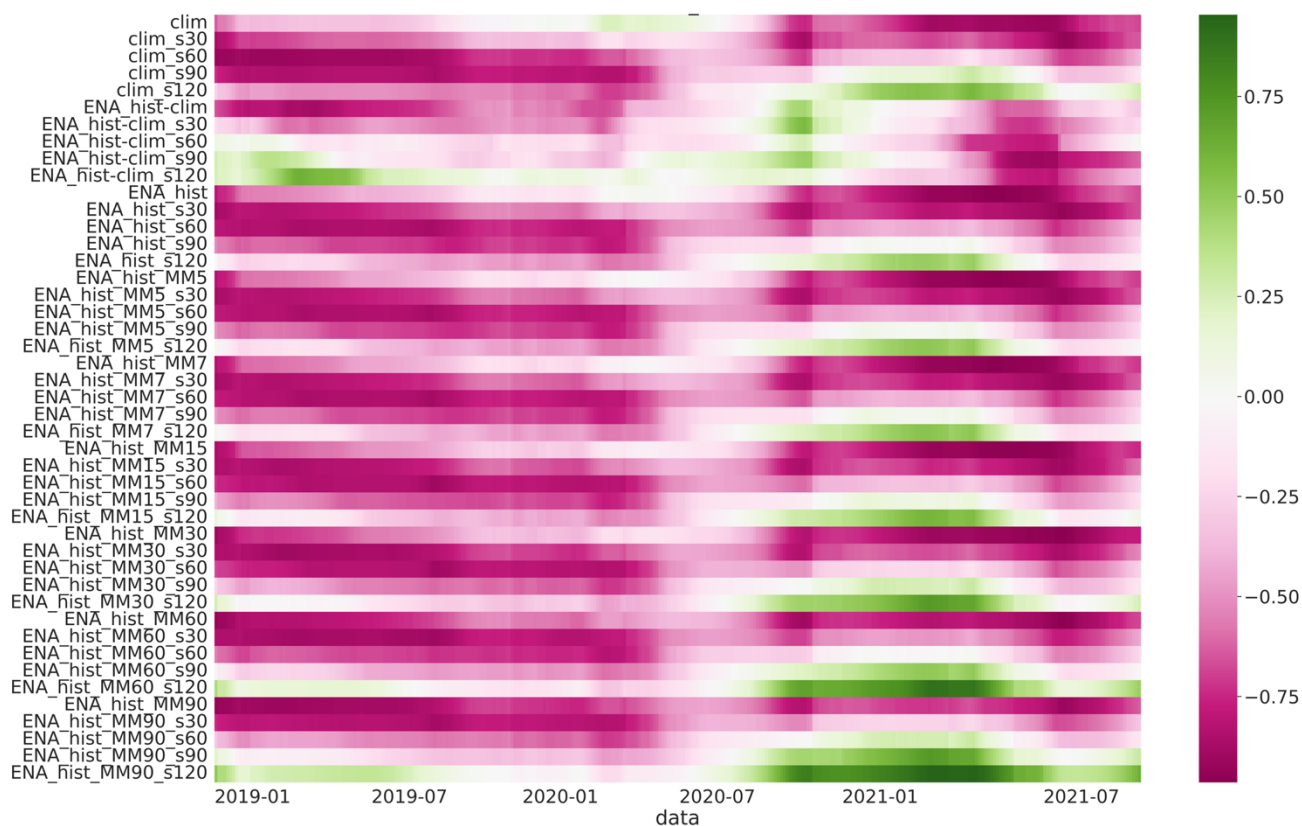


Fig. 14. Spearman correlation for the Marginal Cost of Operation (CMO) feature of the North subsystem.

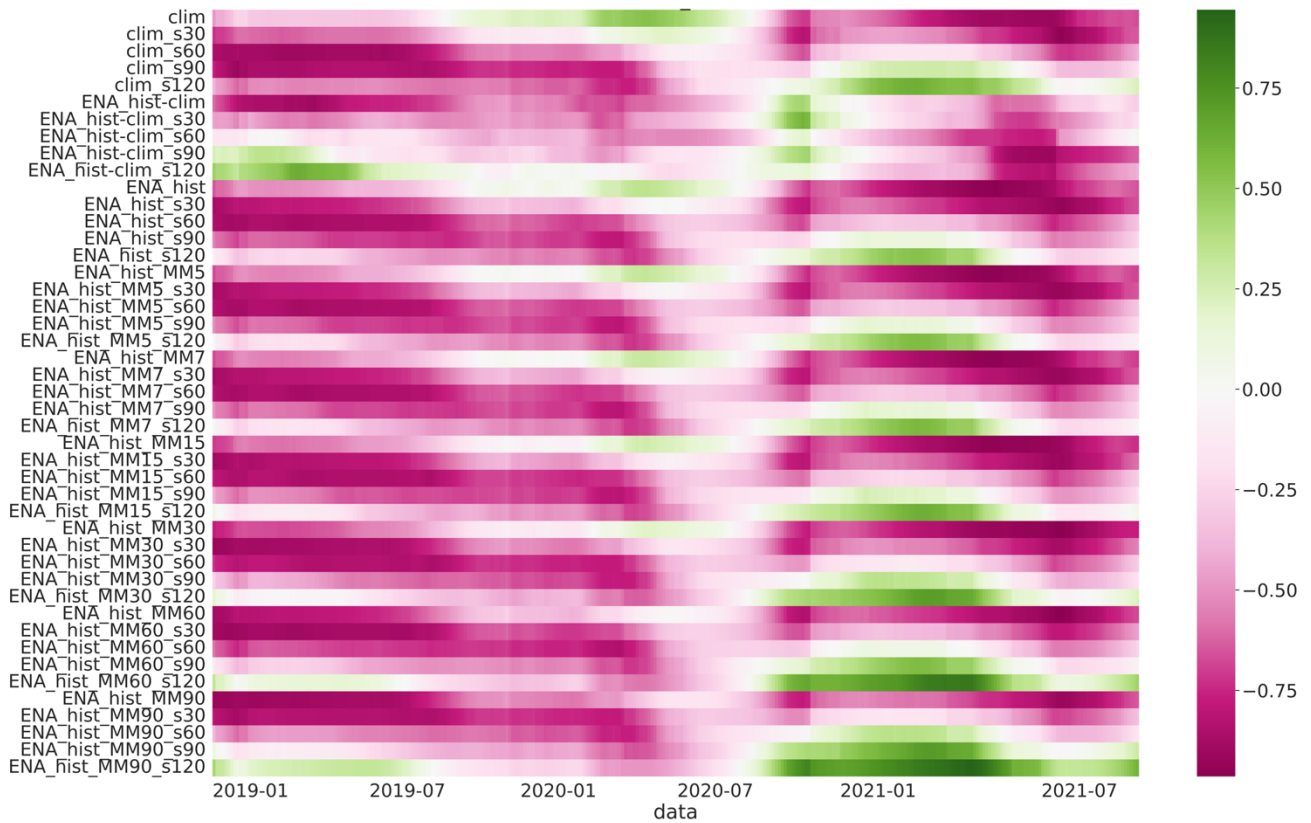


Fig. 15. Spearman correlation for the Marginal Cost of Operation (CMO) feature of the Northeast subsystem.

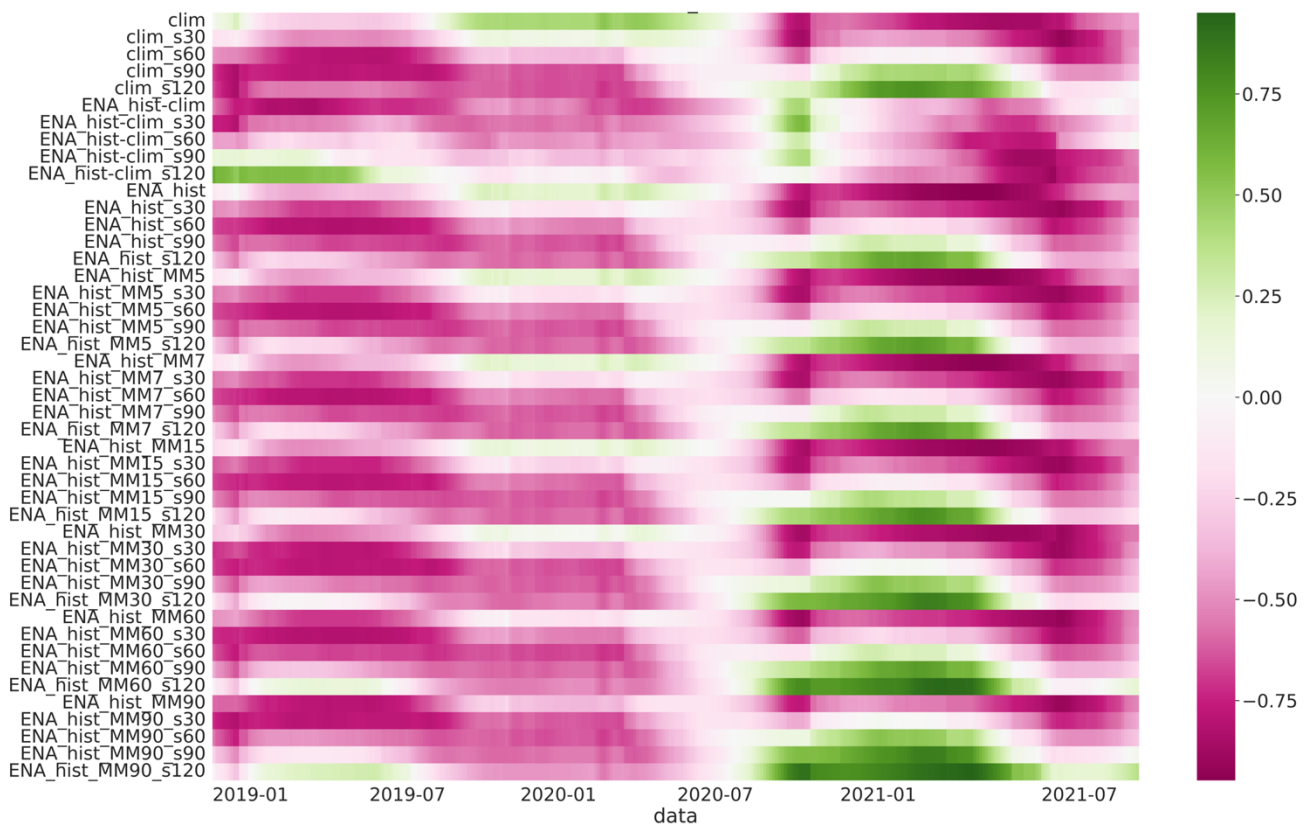


Fig. 16. Spearman correlation for the Marginal Cost of Operation (CMO) feature of the South subsystem.

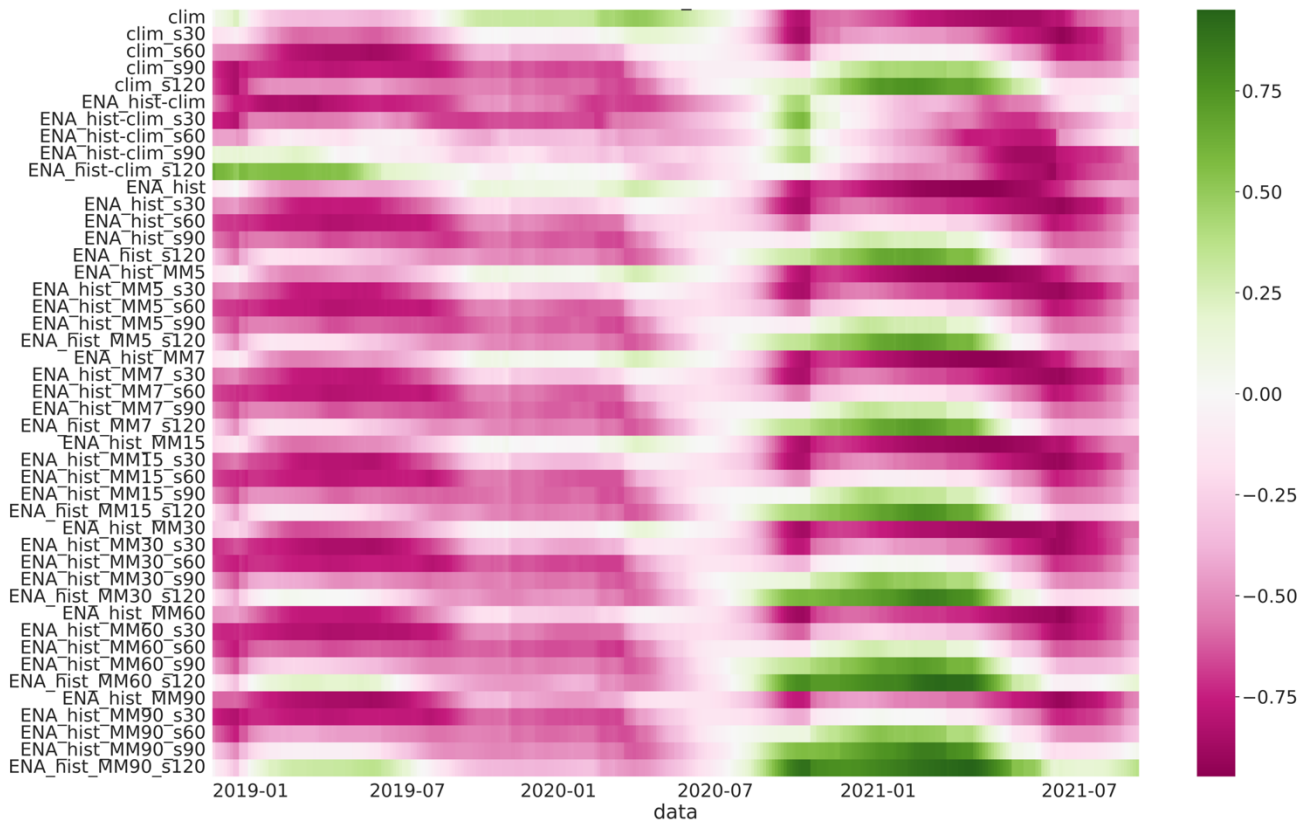


Fig. 17. Spearman correlation for the Marginal Cost of Operation (CMO) feature of the Southeast/Central-West subsystem.

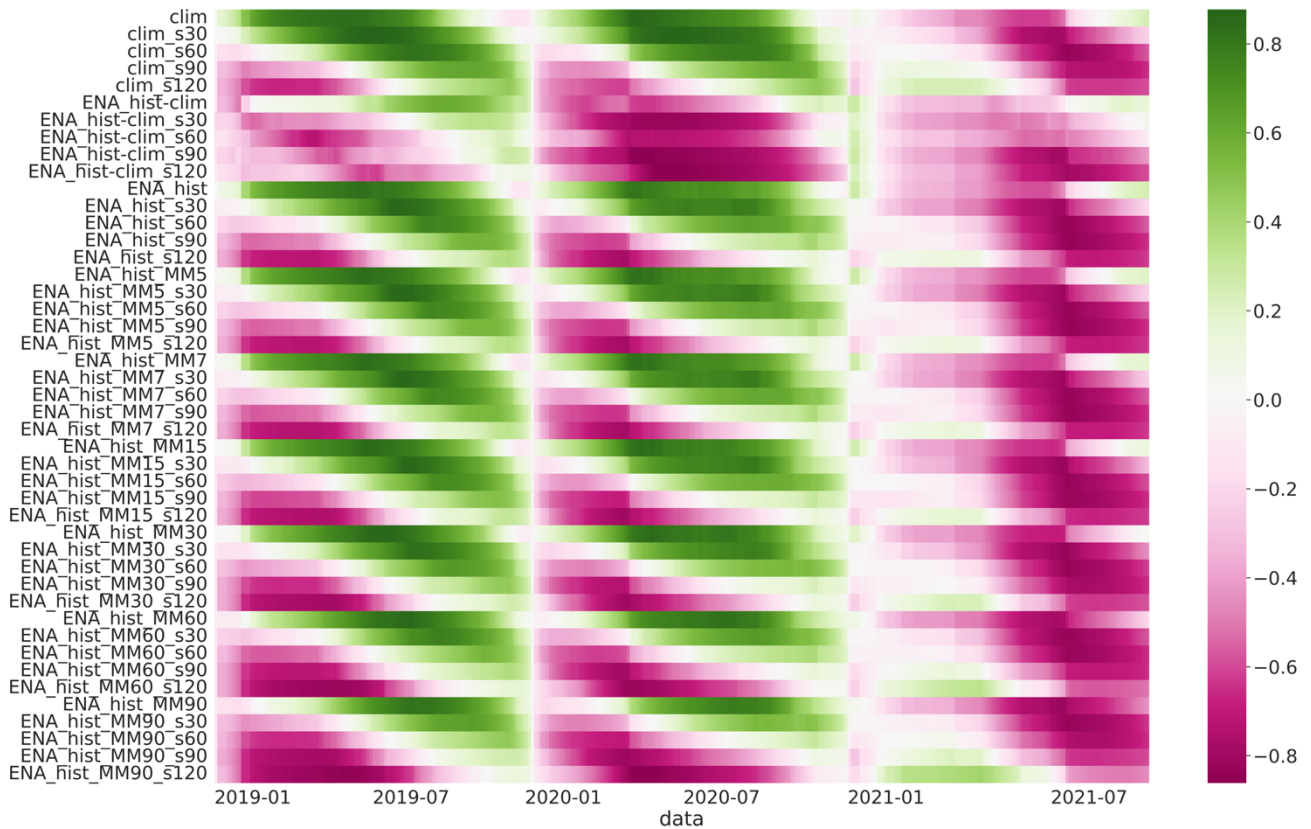


Fig. 18. Spearman correlation for the Gross Domestic Product (GDP) feature.



Fig. 19. Spearman correlation for the Economic Activity Index (IBC) feature.



Fig. 20. Spearman correlation for the Geopolitical Risk Index (GPRD) feature.

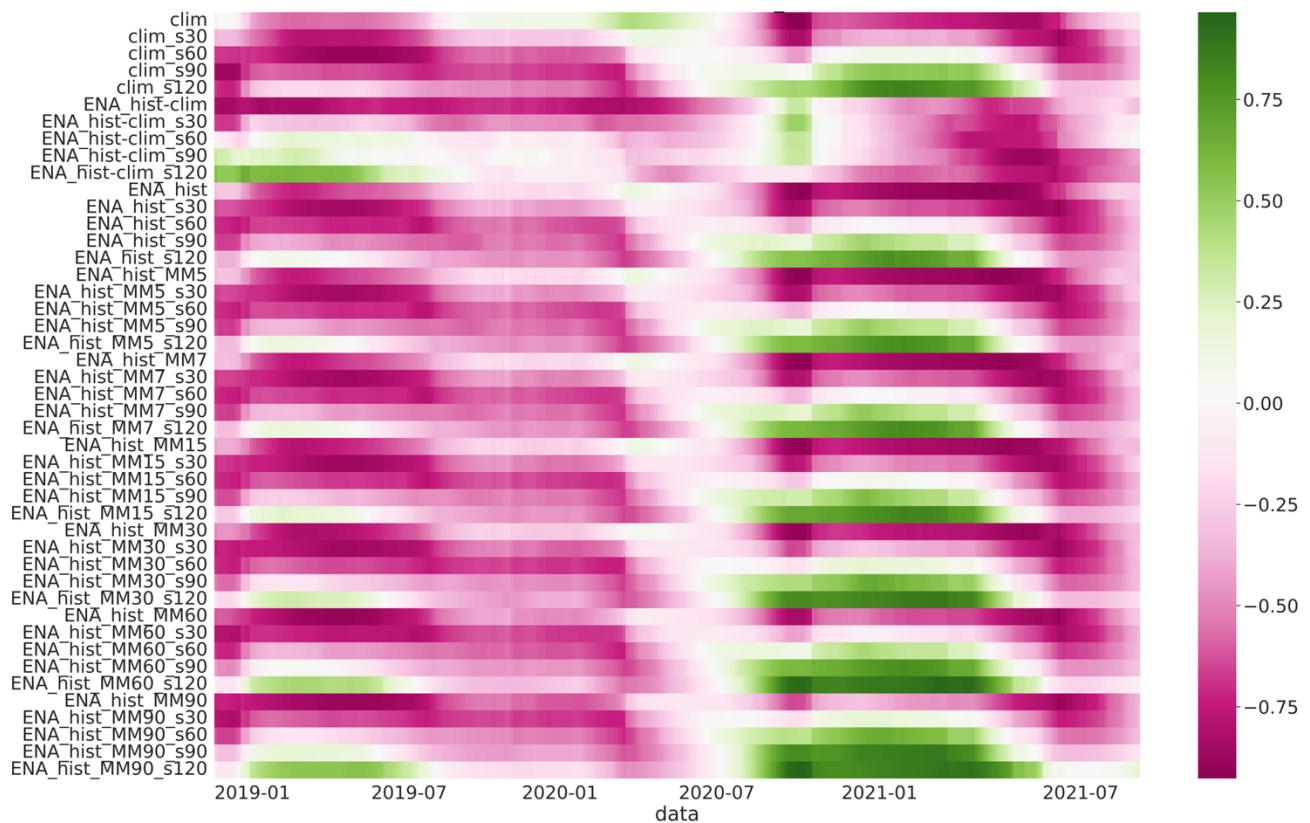


Fig. 21. Spearman correlation for the Volume Weighted Average Price (VWAP) feature for the product M+0.

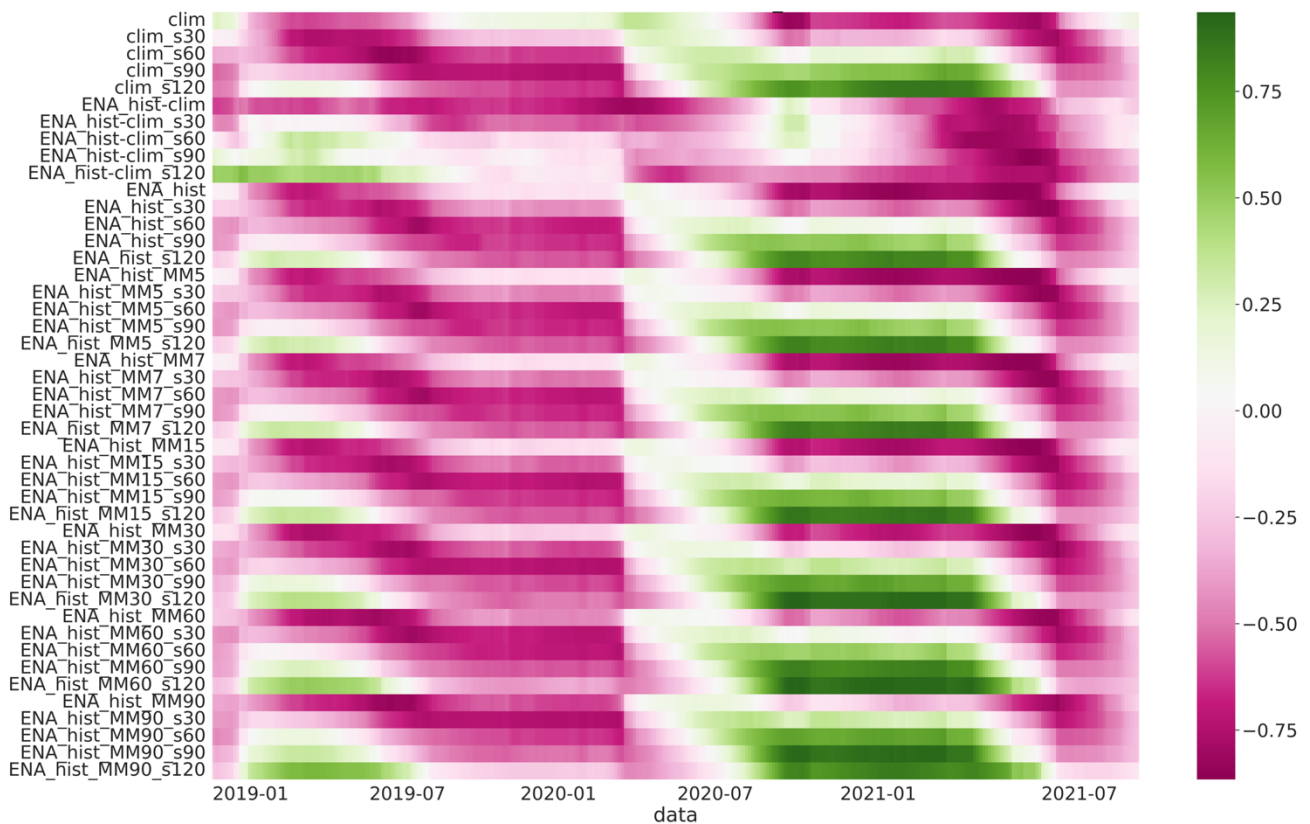


Fig. 22. Spearman correlation for the Volume Weighted Average Price (VWAP) feature for the product M+1.

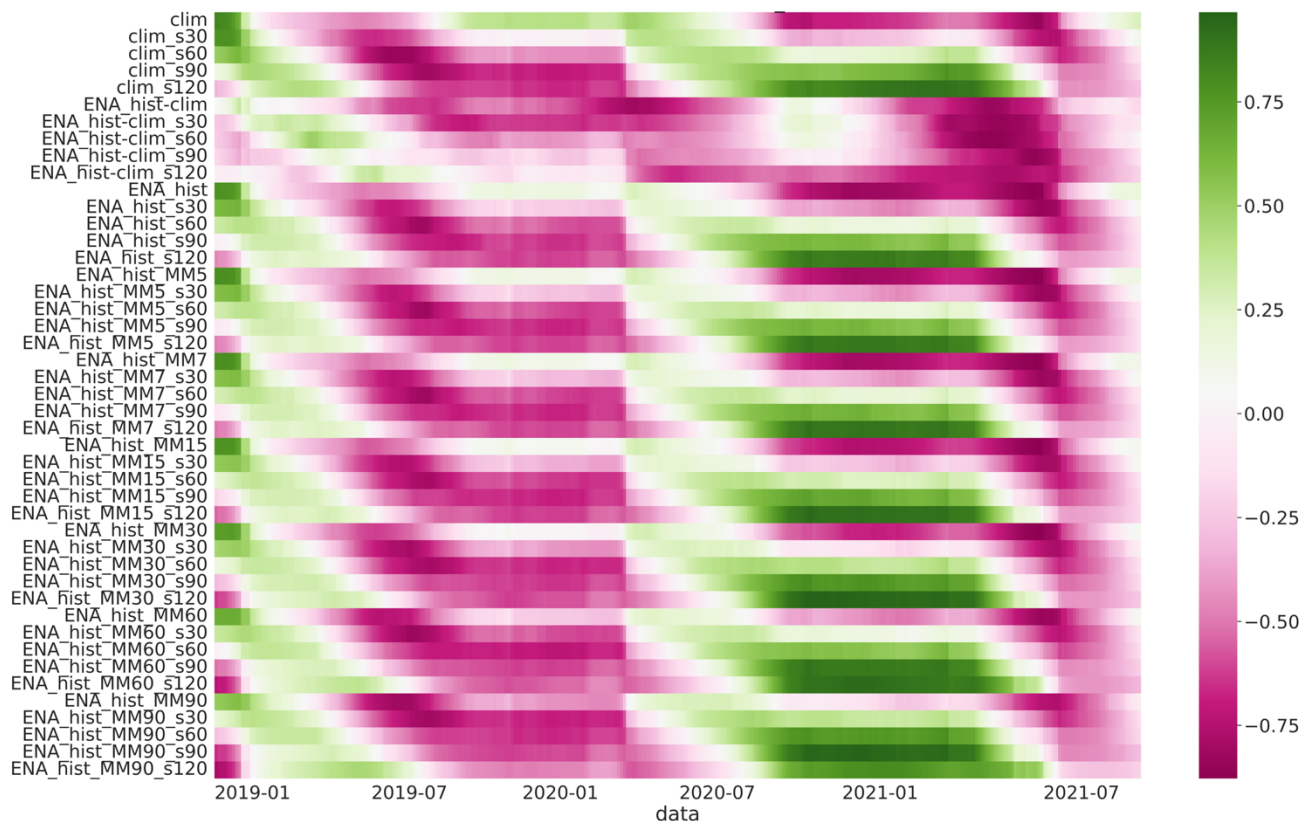


Fig. 23. Spearman correlation for the Volume Weighted Average Price (VWAP) feature for the product M+2.

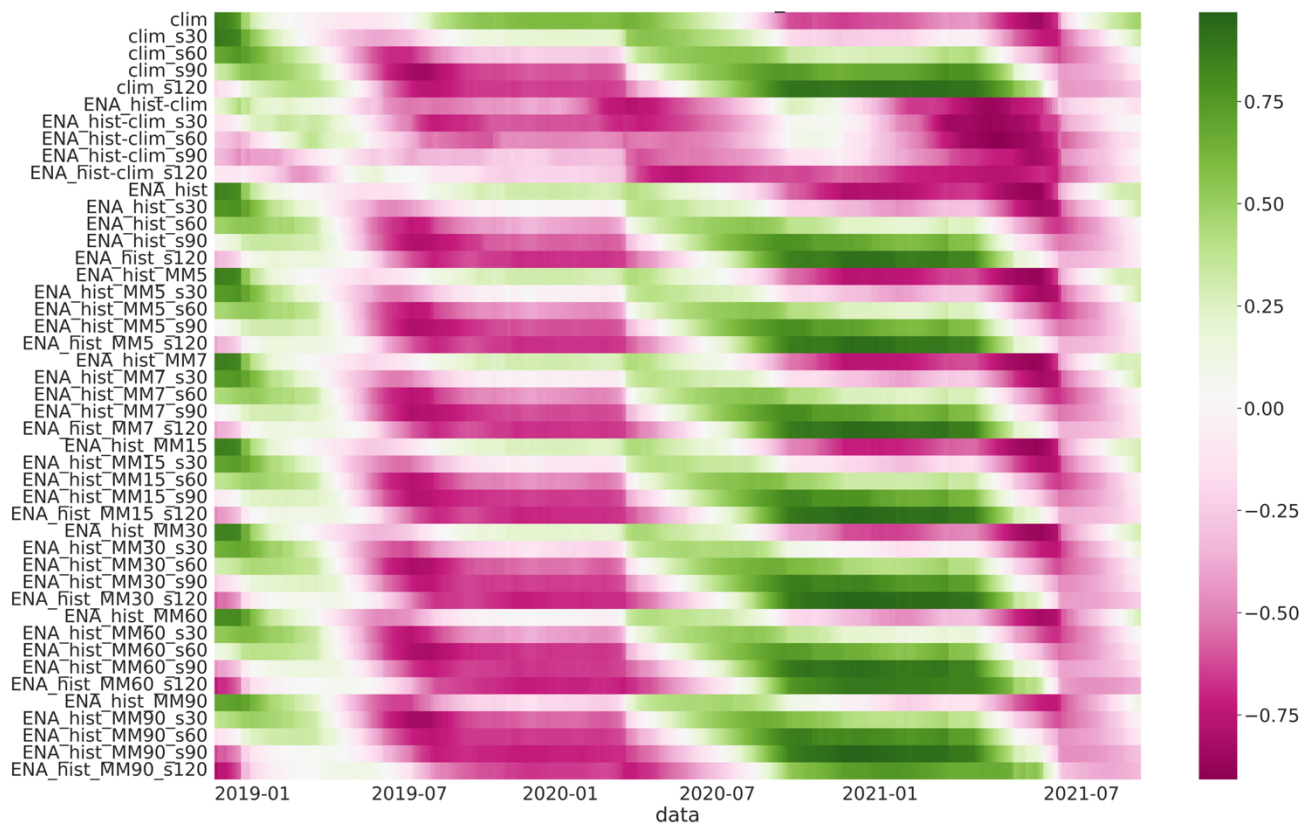


Fig. 24. Spearman correlation for the Volume Weighted Average Price (VWAP) feature for the product M+3.

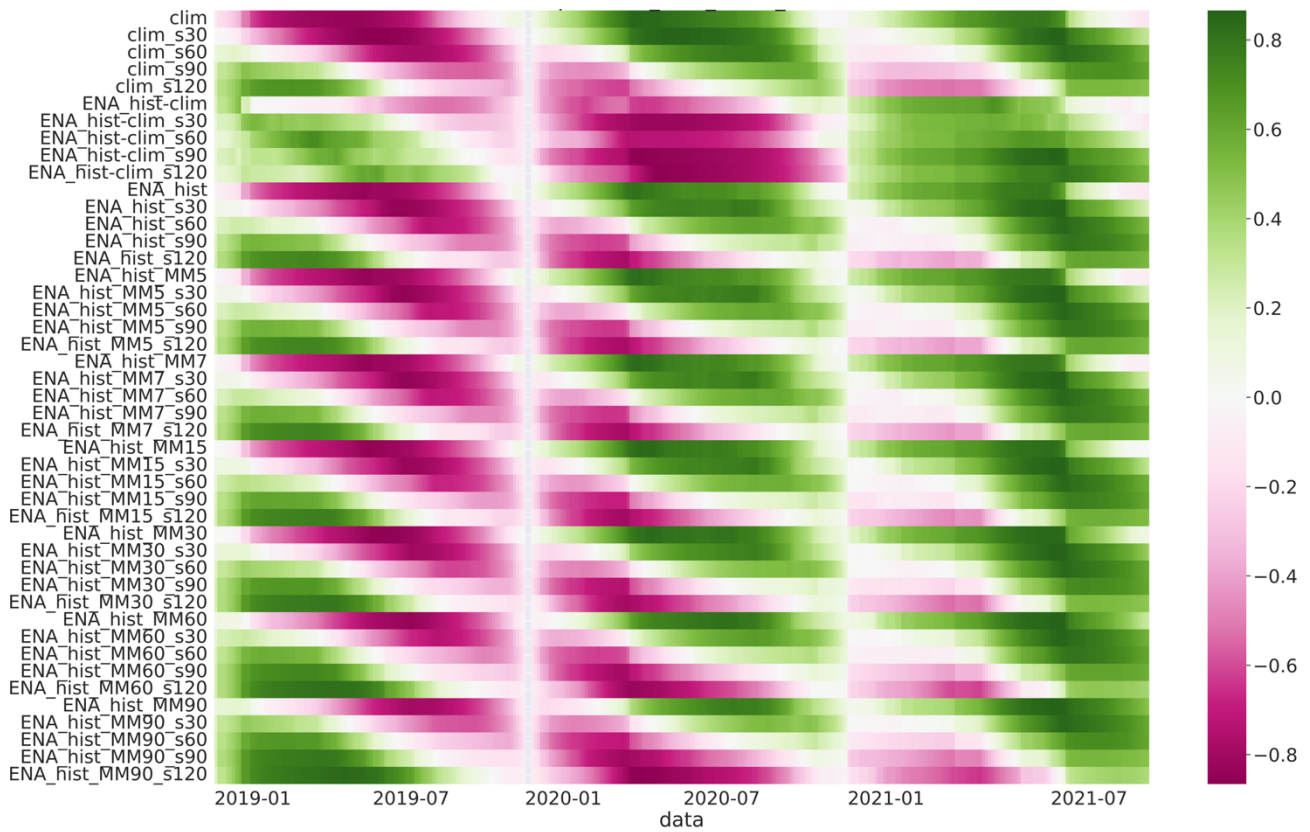


Fig. 25. Spearman correlation for the Total capacity of SIN transmission lines feature.

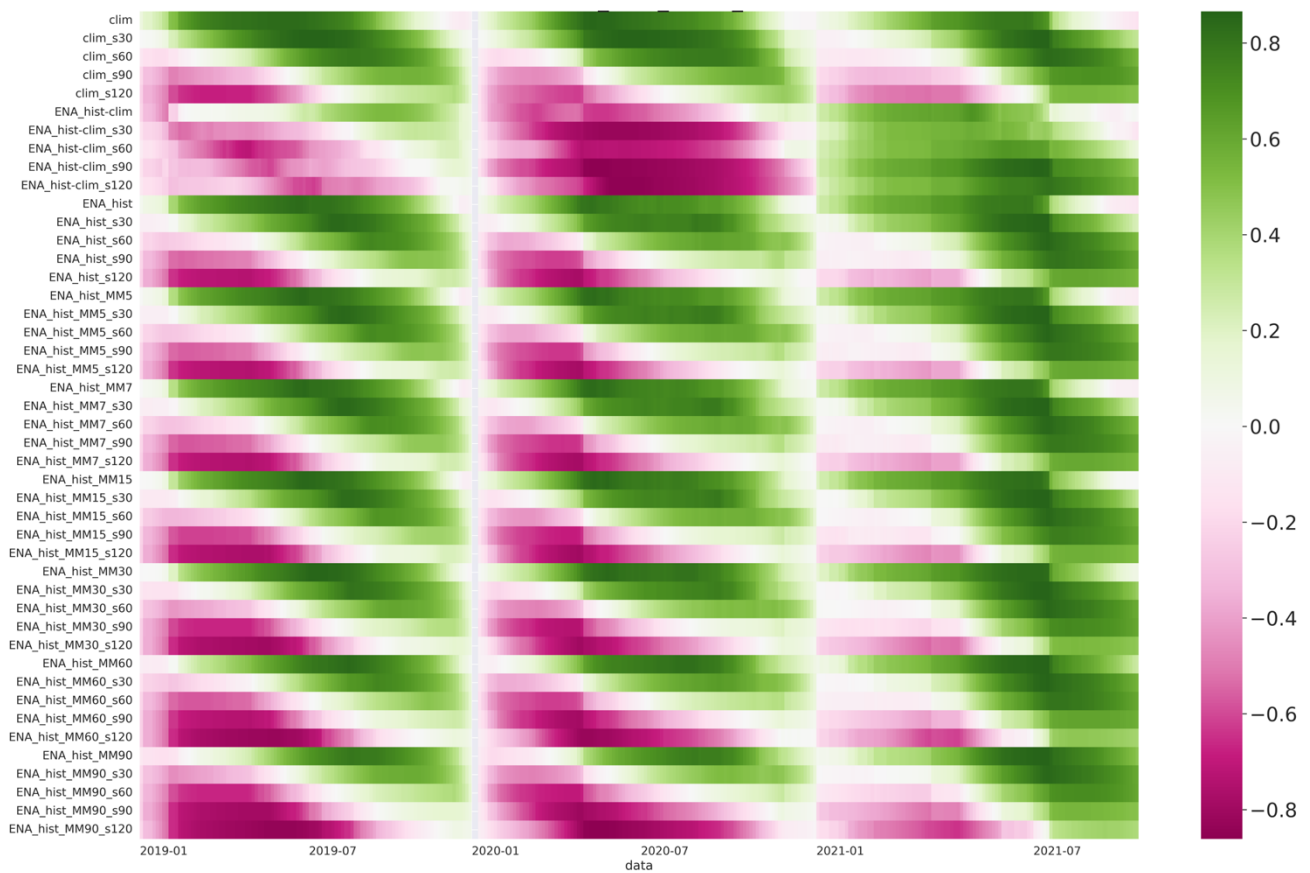


Fig. 26. Spearman correlation for the Total size of SIN transmission lines feature.

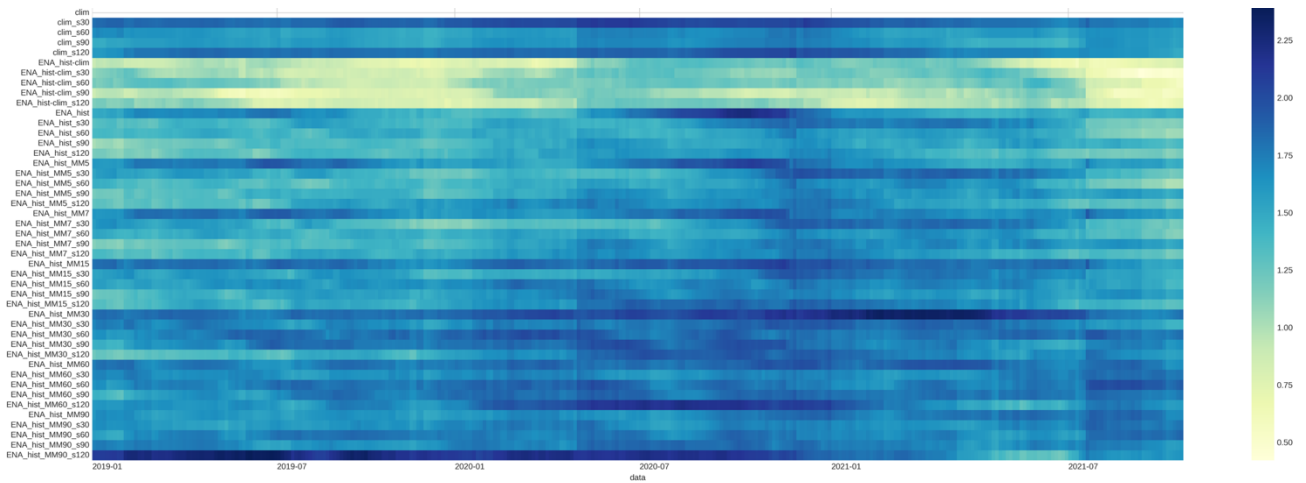


Fig. 27. Mutual Information for the Climatology (Clim) feature.

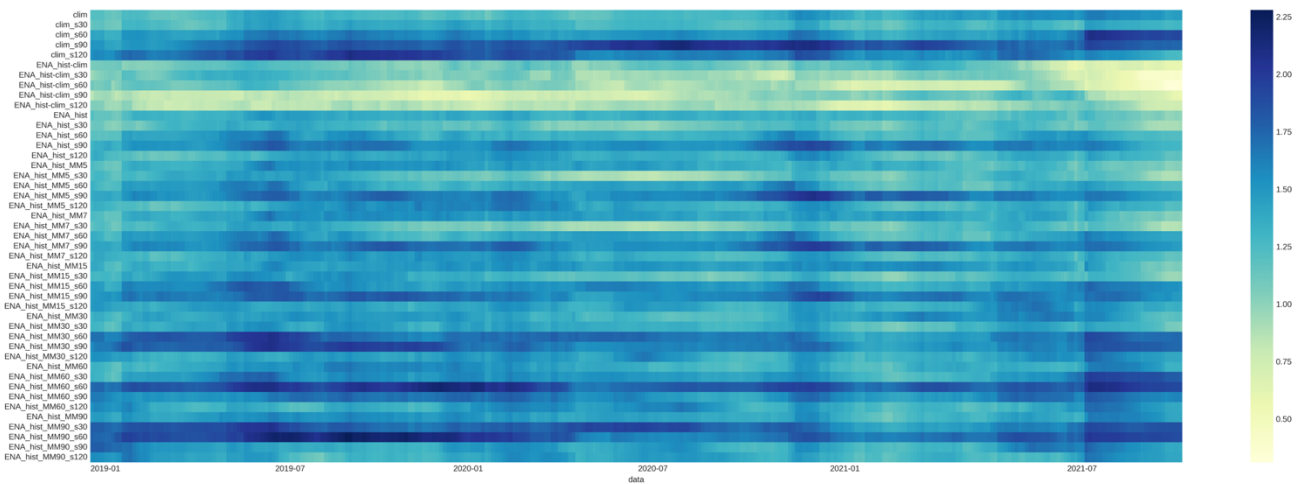


Fig. 28. Mutual Information for the Average Stored Energy (EARm) feature of the North subsystem.

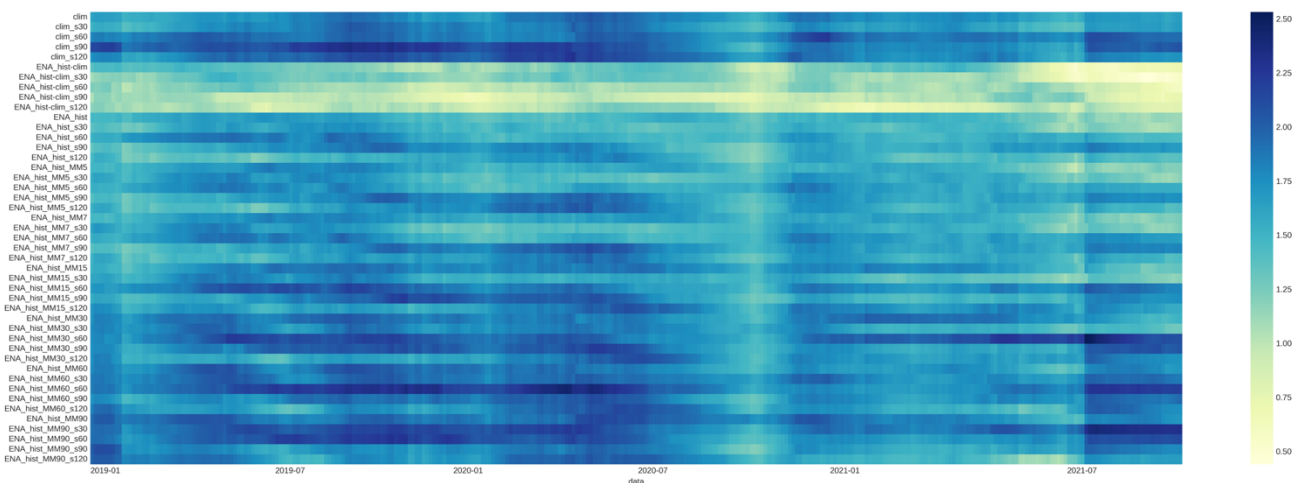


Fig. 29. Mutual Information for the Average Stored Energy (EARm) feature of the Northeast subsystem.

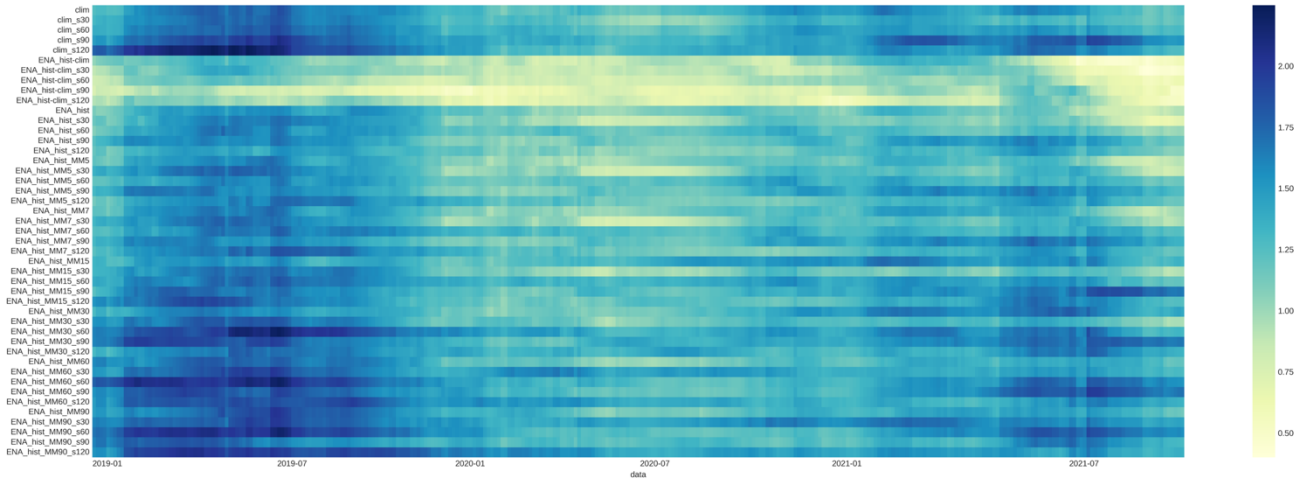


Fig. 30. Mutual Information for the Average Stored Energy (EARm) feature of the South subsystem.

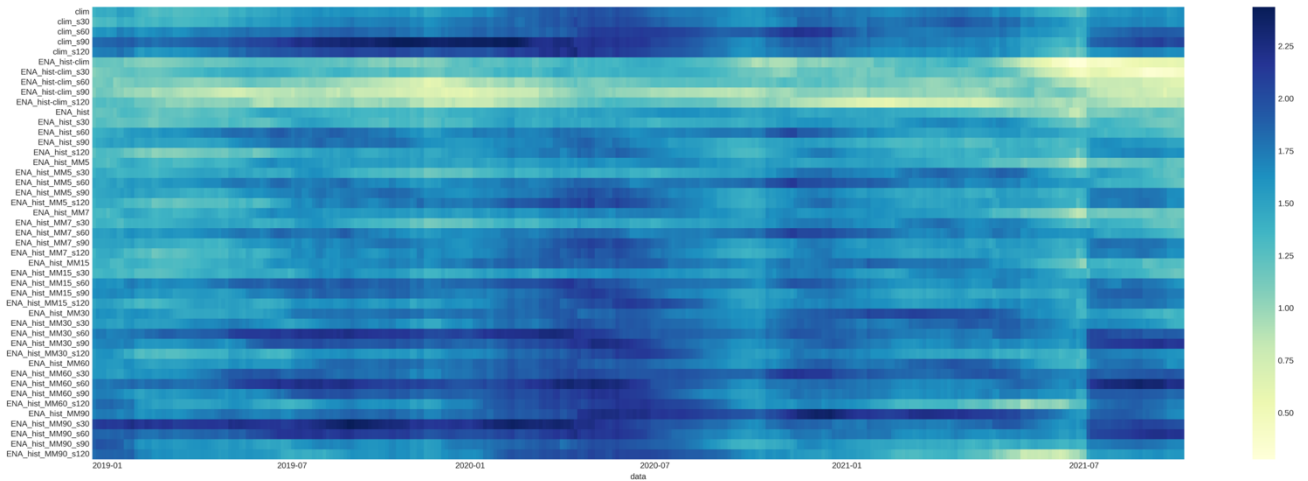


Fig. 31. Mutual Information for the Average Stored Energy (EARm) feature of the Southeast/Central-West subsystem.

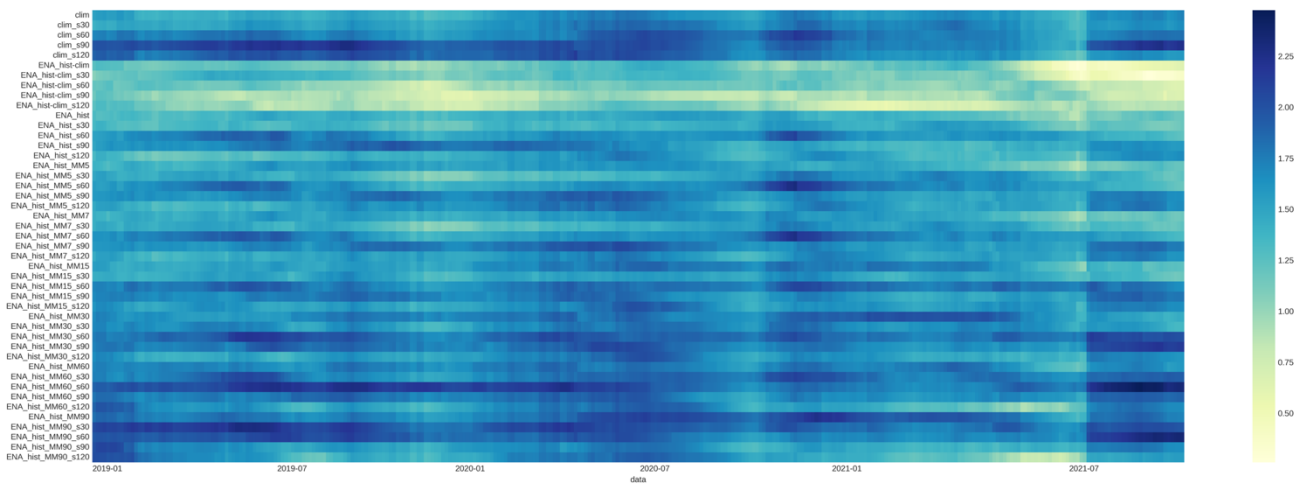


Fig. 32. Mutual Information for the Average Stored Energy (EARm) feature of the National Interconnected System (SIN).

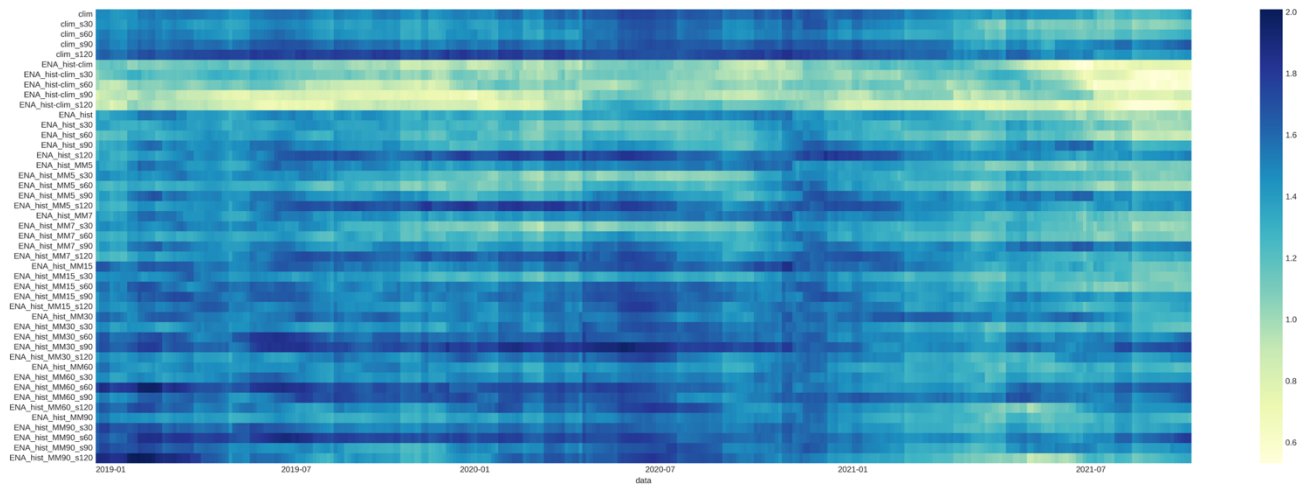


Fig. 36. Mutual Information for the Electricity Load Projection feature or the product M+1.

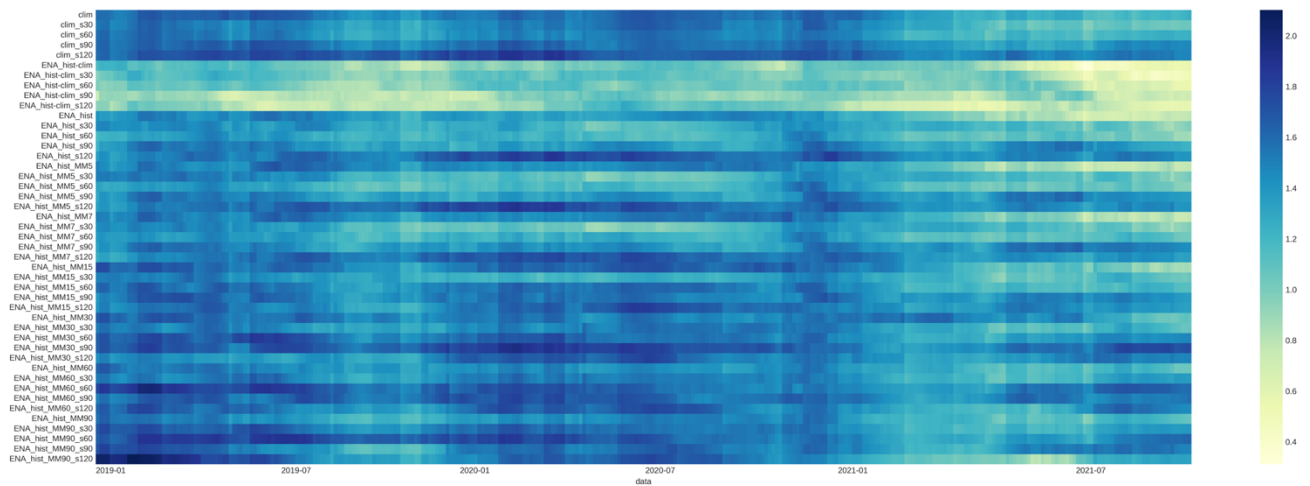


Fig. 37. Mutual Information for the Electricity Load Projection feature for the product M+2.

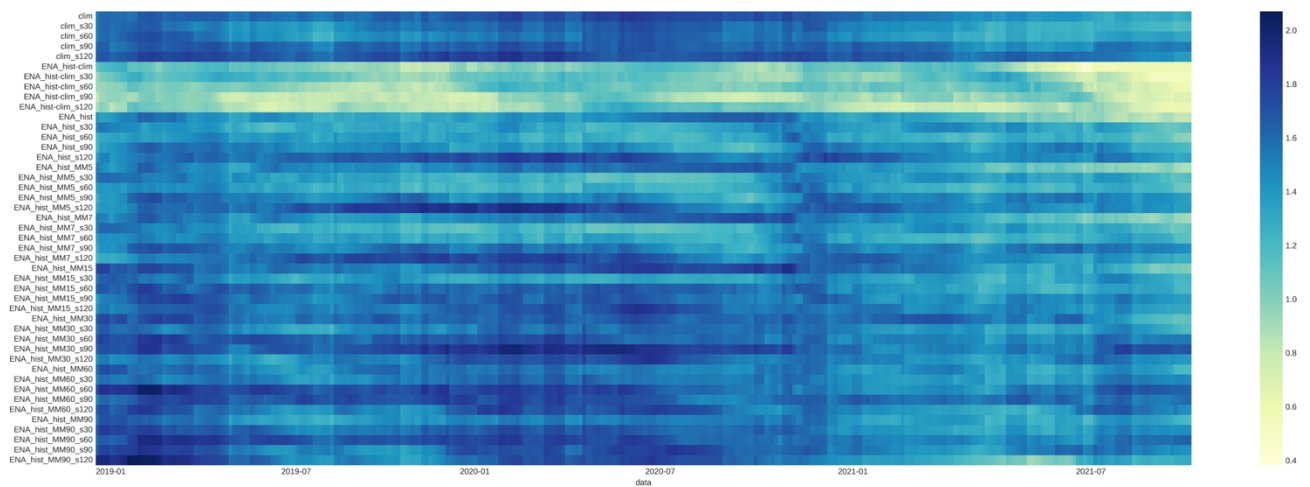


Fig. 38. Mutual Information for the Electricity Load Projection feature for the product M+3.

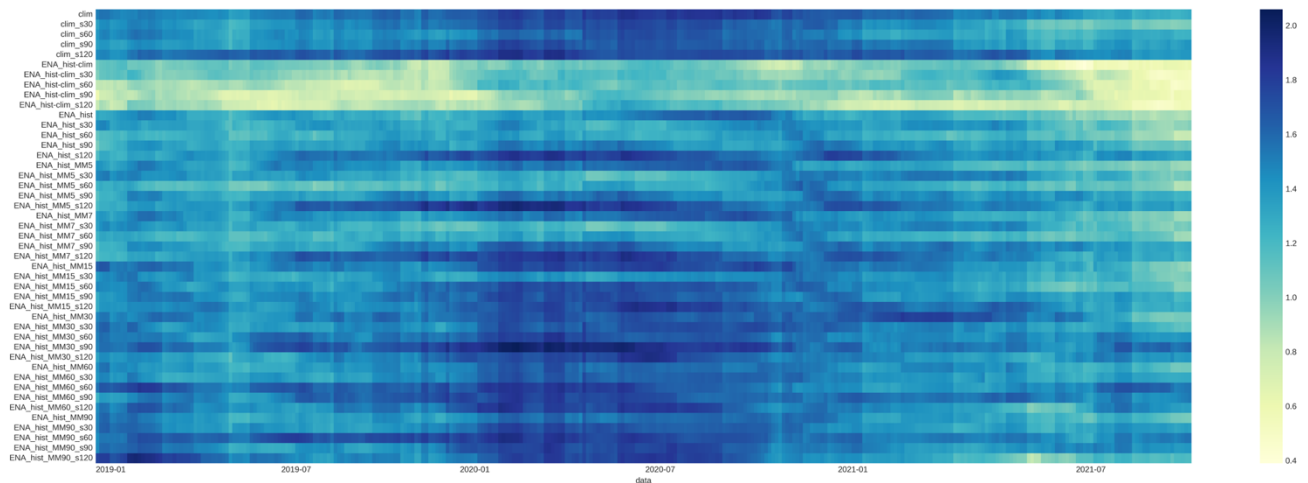


Fig. 39. Mutual Information for the Electricity Load Projection feature for the product M+4.

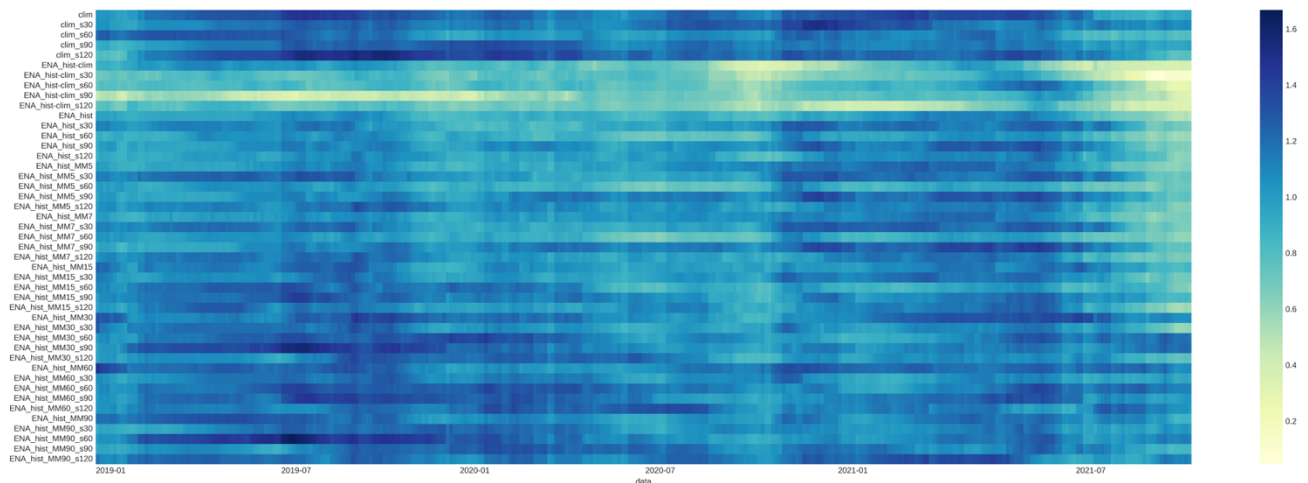


Fig. 40. Mutual Information for the Marginal Cost of Operation (CMO) feature of the North subsystem.

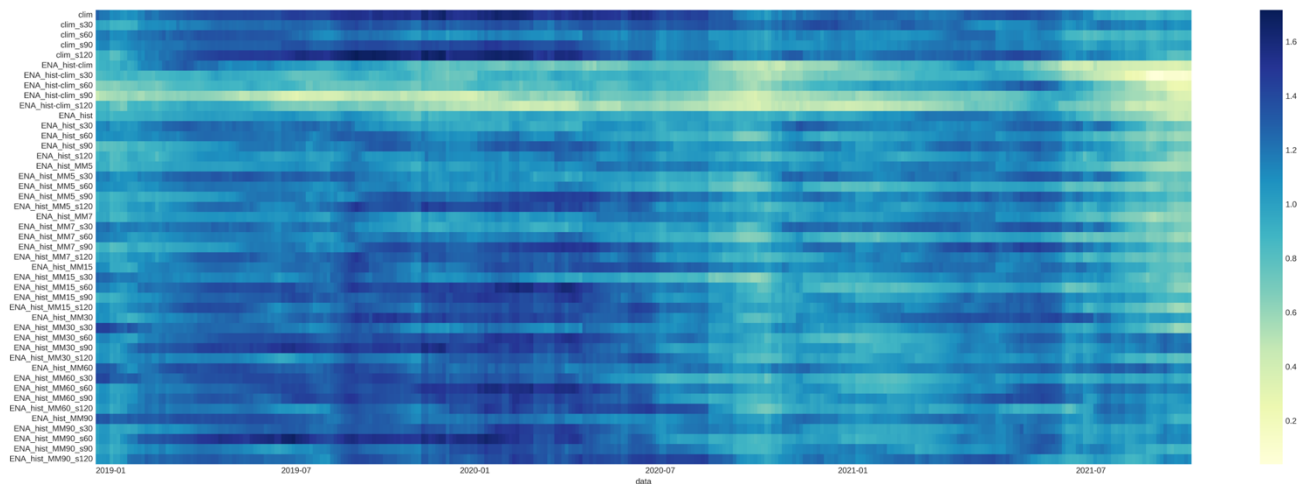


Fig. 41. Mutual Information for the Marginal Cost of Operation (CMO) feature of the Northeast subsystem.

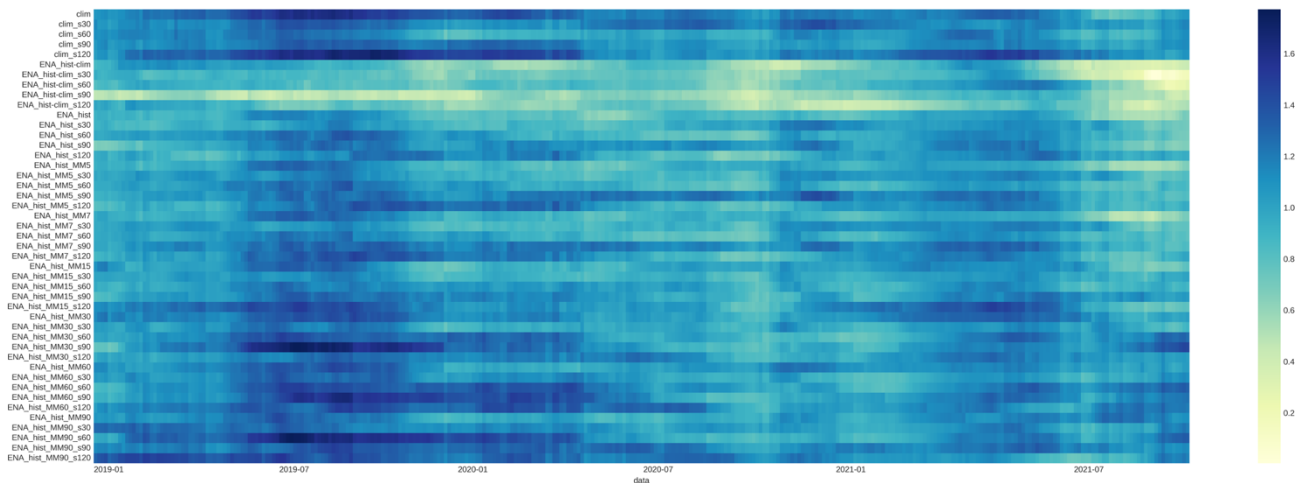


Fig. 42. Mutual Information for the Marginal Cost of Operation (CMO) feature of the South subsystem.

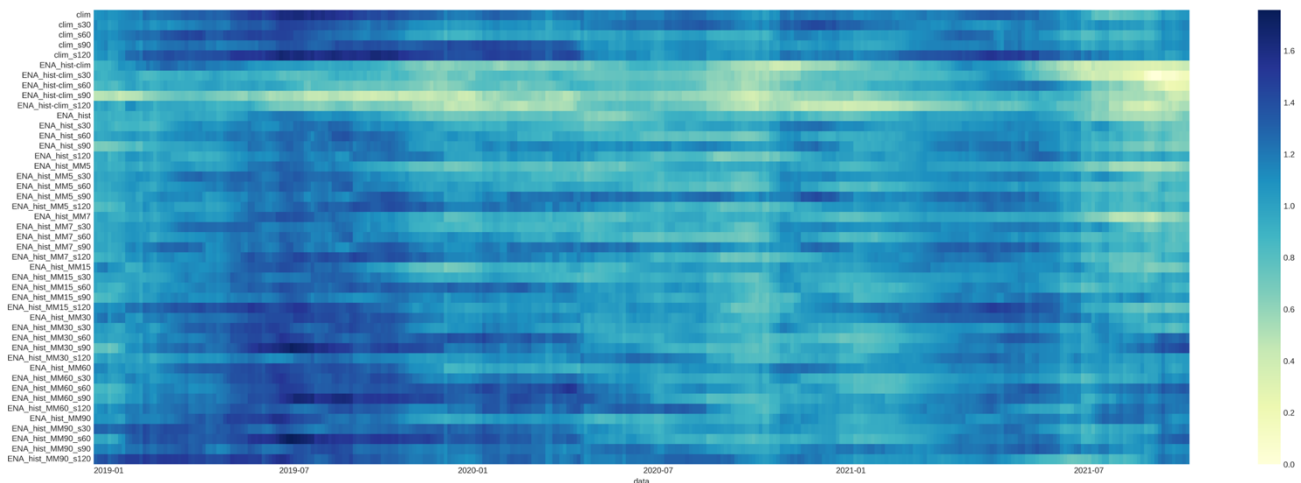


Fig. 43. Mutual Information for the Marginal Cost of Operation (CMO) feature of the Southeast/Central-West subsystem.

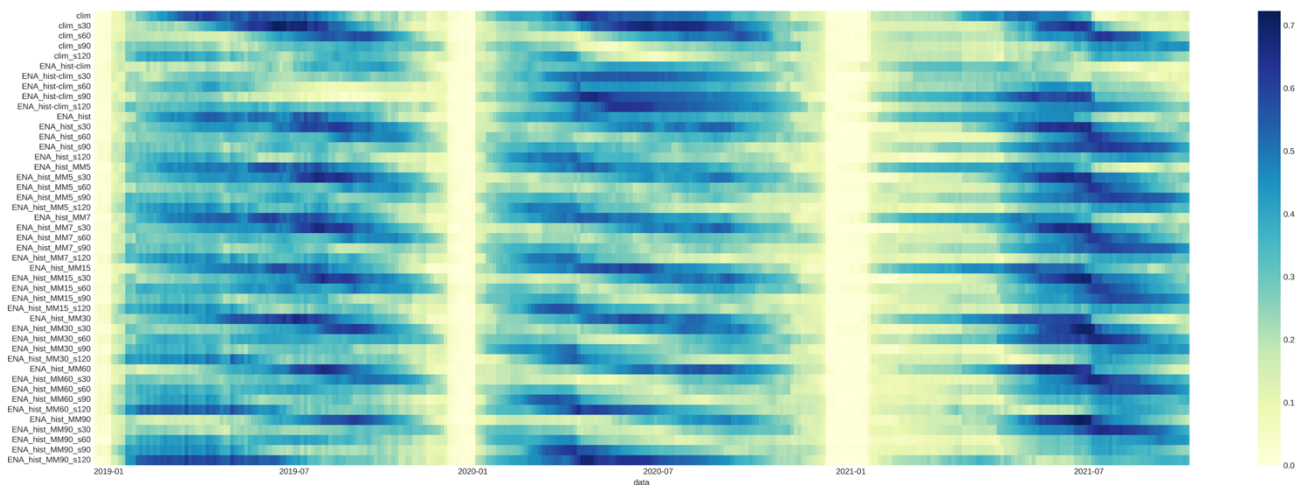


Fig. 44. Mutual Information for the Gross Domestic Product (GDP) feature.

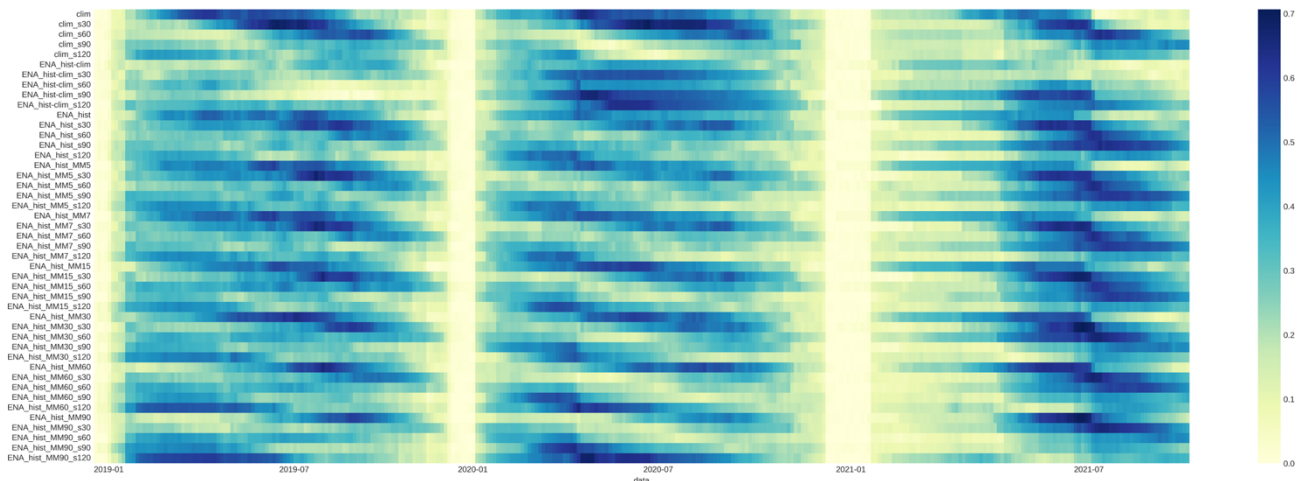


Fig. 45. Mutual Information for the Economic Activity Index (IBC) feature.



Fig. 46. Mutual Information for the Geopolitical Risk Index (GPRD) feature.

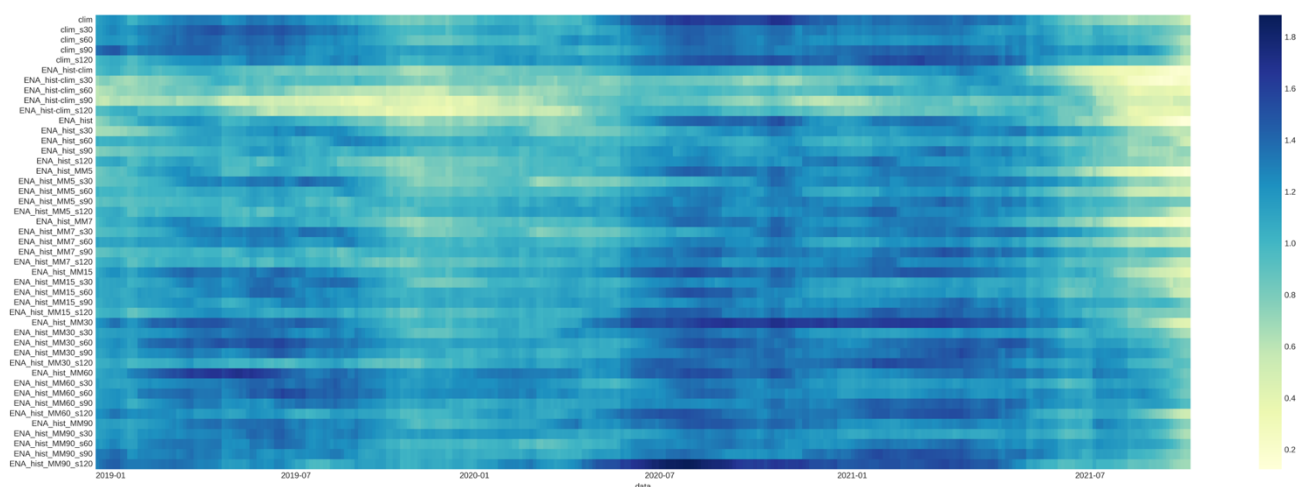


Fig. 47. Mutual Information for the Volume Weighted Average Price (VWAP) feature for the product M+0.

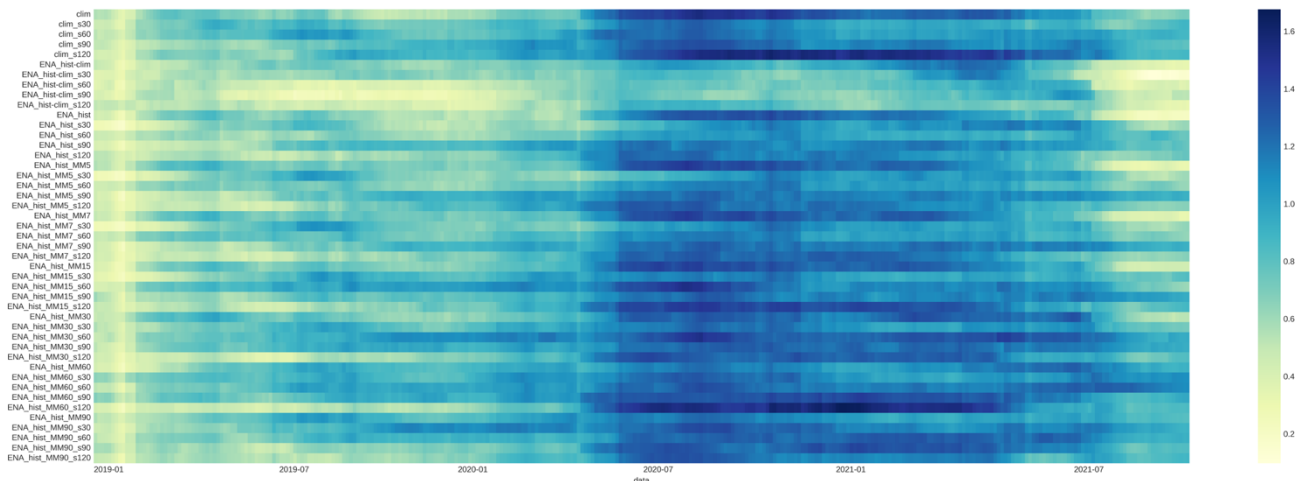


Fig. 48. Mutual Information for the Volume Weighted Average Price (VWAP) feature for the product M+1.

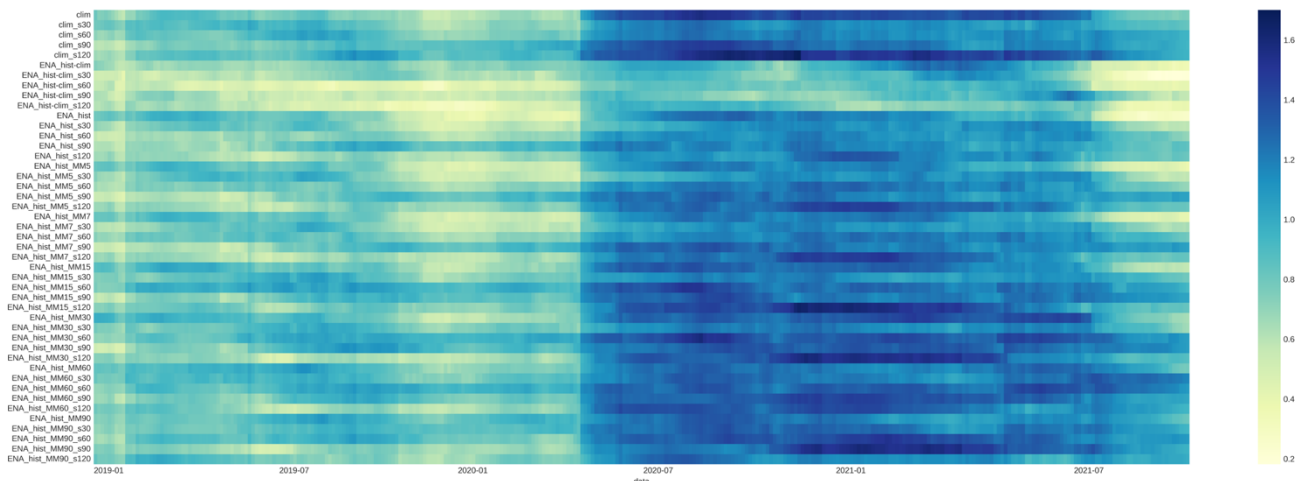


Fig. 49. Mutual Information for the Volume Weighted Average Price (VWAP) feature for the product M+2.

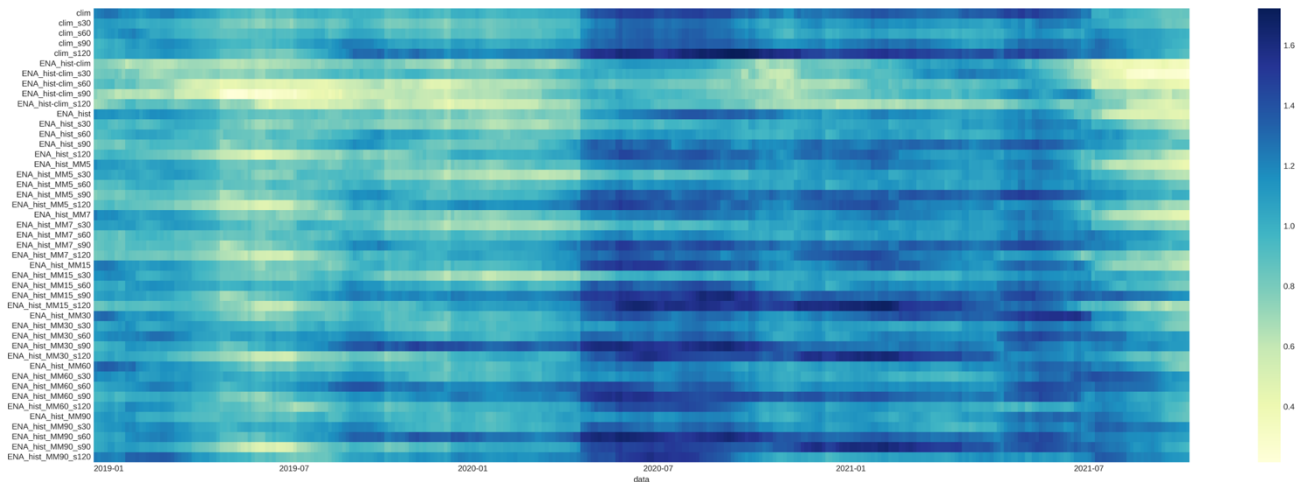


Fig. 50. Mutual Information for the Volume Weighted Average Price (VWAP) feature for the product M+3.

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